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A Two-Stage Hybrid Feature Selection and Ensemble Learning Framework With Explainable AI for Accurate PCOS Prediction

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ABSTRACT

Background and Aims: Polycystic ovary syndrome (PCOS) is the most prevalent endocrine ailment impacting women of reproductive age, distinguished by ongoing imbalances of hormones that lead to the growth of various ovarian cysts alongside other health issues. Infrequent or delayed menstrual periods, and also frequently excessive amounts of androgen, a male hormone, are characteristics of women with PCOS. Despite other consequences, this disease may lead to weight gain, undesirable body hair, type 2 diabetes, and gestational diabetes. The number of cases of PCOS has shockingly ascended in the past few years, according to statistics. But the clinical diagnosis procedure for PCOS in the real world becomes critical since the precision of interpretations greatly relies on the knowledge of the doctor. Therefore, a PCOS prediction model using artificial intelligence might be an appropriate supplement to the lengthy and error-prone diagnosis method.

Methods: To strictly prevent data leakage, the dataset underwent an 80/20 train-test split before applying SMOTE and two-stage feature selection exclusively to the training data. This framework combined Chi-Square filtering with Recursive Feature Elimination to identify the most pertinent features. A hybrid XGBoost (XGB) and Multi-Layer Perceptron (MLP) ensemble was selected from multiple machine learning models following rigorous evaluation using 10-fold stratified cross-validation. Finally, SHapley Additive exPlanations and Local Interpretable Model-Agnostic Explanations were integrated to significantly enhance overall model transparency.

Results: The hybrid ensemble model (XGB + MLP) outperformed conventional single ML models, with the greatest classification accuracy of 96.33% among all evaluated models. The model's interpretability and performance were greatly enhanced by the 16 characteristics that were chosen.

Conclusion: Combining hybrid ensemble learning, two-stage feature selection, and XAI approaches provides a computationally efficient, dependable, and interpretable method for PCOS diagnosis. Practitioners may find this model to be a useful decision-support tool that improves the accuracy of diagnosis and lessens the need for human interpretation.

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1 | Introduction

PCOS serves as the typical abbreviation for polycystic ovary syndrome. PCOS is among the most prevalent endocrinological disorders [1], which generally comes on with a significant rise in the male hormone androgen in the female body. This leads to a long-term interruption in hormonal levels, which, in consequence, affects normal ovarian functions and causes numerous cysts to form inside the ovary [2]. The symptoms of the patient dictate how long PCOS rehabilitation will run. The name of this disorder arises from the cysts that develop on the ovaries of some afflicted individuals; however, they are neither the main cause of the disorder nor typical symptoms. Anovulation with irregular menstruation, oligomenorrhea, amenorrhea, polycystic ovarian morphology, and hyperandrogenism with acne, hirsutism, and alopecia are just some of the warning signs that may be used to forecast this varied and heterogeneous disorder [3]. Inflammation, heredity, or insulin resistance are some possible reasons for PCOS, while the precise explanation is yet unclear. PCOS increased the risk of depression, diabetes, high blood pressure, and other ailments [4]. Epidemiological research [5] demonstrates that PCOS drives a number of harmful, life-threatening effects that are common among PCOS patients, impacting one in ten premenopausal reproductive women globally and causing 44–70% of women to suffer from numerous critical adverse effects. This condition is connected with many different metabolic and psychological disorders that shrinking the quality of life that is healthier. Ovarian and endometrial cancers are more common in women with PCOS, and both may be deadly if not recognized in the early stages [6]. Therefore, an extensive number of symptoms and disorders are linked to PCOS, the most common endocrinological ailments. Stein and Leventhal found in 1935 that 5%–10% of women between the ages of 12 and 45 had PCOS [7]. Over 5 million women worldwide suffer from these issues, and 70% of these cases remain untreated. Ethnicity may have an impact when it comes to diagnosis rates, which are 4.8% for white women, 8% for African American women, 6.8% for Spanish women, and 31.3% for Asian women [8]. Even though the exact cause of PCOS is still unknown, insulin insensitivity seems to play a part, as demonstrated by the darkening of particular areas of the skin. Furthermore, PCOS may also be brought on by lifestyle factors, including food and inactivity [9]. Symptom management is the main goal of treatment: anti-androgens prevent acne and excessive hair growth; diabetes meds like metformin control insulin and periods; therapies for fertility encourage ovulation; and hormonal birth control regulates cycles and lowers testosterone. Laparoscopic ovarian drilling may lower testosterone production as well as repair ovulation if medication is inadequate [10]. Based on recent research, if a well-standardized diagnosis technique is used to identify PCOS early on, the disease may be treated with appropriate symptom-oriented, long-term, dynamic drugs together with an optimal diet. However, doctors find it exceedingly difficult to accurately recognize PCOS in its earliest phases due to the vast range of symptoms linked to the ailment and the emergence of other concurrent gynecological issues. Furthermore, a number of clinical test assessments by trained healthcare professionals are essential for the accurate diagnosis of PCOS, which is sometimes impossible in places with a lack of resources and skilled medical professionals. Many young women are suffering the negative effects of this debilitating illness, particularly in rural areas in underdeveloped and least developed nations, as a result of going undiagnosed and untreated because they have little other choice than to see qualified clinicians.

Complex ailments are being identified as well as treated with the use of artificial intelligence (AI). PCOS can possibly be identified sooner, before it worsens, by using machine learning techniques. Our proposed study suggests a powerful, mathematically based machine learning system that can be used as a clinical decision-support system to enable the early and accurate prediction of PCOS. The framework is based on the comprehensive tabular data that contains biological and clinical indicators. To ensure data integrity, the pipeline incorporates rigorous preprocessing, including missing value imputation, categorical encoding, and MinMax scaling. A two-stage hybrid feature selection method is used to maximize the predictive performance. The first step in this involves Chi-Square filtering to reduce the feature space, and then Recursive Feature Elimination (RFE) to identify a small set of 16 critical features that is highly stable.

There was an evaluation of a wide range of machine learning classifiers, such as Random Forest (RF), K-Nearest Neighbors (KNN), SVM, Logistic Regression (LR), and different gradient boosting algorithms. Moreover, many combinations of votes in the ensembles were tried to obtain the maximum classification accuracy. The last suggested ensemble that integrated Extreme Gradient Boosting (XGB) and Multi-Layer Perceptron (MLP) had a remarkable accuracy of 96.33%. Lastly, to guarantee clinical trust and model transparency, Explainable AI (XAI) methods (SHapley Additive exPlanations [SHAP] and Local Interpretable Model-Agnostic Explanations [LIME]) were combined to offer a global feature importance and instance-specific, localized explanations.

Although several studies have investigated feature selection methods and ensemble learning for predicting PCOS, they are unstable on small clinical data or do not provide reliable generalization measures. To address these research gaps and take it to the next level, the contributions of this study are:

- The study designed a multi-wrapper consensus strategy that cross-references four diverse models (CatBoost [CB], LR, XGB, and RF) to extract a highly stable, model-agnostic feature set and conducted the first systematic comparison within RFE for PCOS to empirically demonstrate CB as the superior wrapper.
- To ensure the model generalizes reliably to unseen data, we directly addressed the instability issues common in small clinical datasets. We confirmed this robustness by calculating specific overfitting margins, assessing learning curve convergence, and running a 10-fold stratified cross-validation (CV).
- Instead of solely optimizing for peak accuracy, the final ensemble was constructed by pairing complementary architectures: a tree-based model (XGB) and a neural network (MLP). To unpack how this complex hybrid model makes decisions, both SHAP and LIME were utilized to achieve deep, dual-level interpretability, moving beyond standard single-model explanations.

- The research bridged the gap between machine learning and clinical relevance by statistically validating the selected features using regression analysis, reporting exact effect sizes and *p*-values to prove the real-world clinical meaning of the algorithmic choices.
- A formal algorithmic breakdown of the computational efficiency of the proposed two-stage selection pipeline is provided, demonstrating its suitability and efficiency for practical clinical deployment.

2 | Related Work

The amount of biological and medical data is growing rapidly. AI and machine learning techniques are being used more and more to comprehend this vast and complicated amount of data. Using clinical factors and vital signs as input data, several researchers have developed AI-driven algorithms for PCOS diagnosis in recent years.

Hassan and Mirza [11] proposed a model for PCOS patient identification. In order to identify PCOS patients, researchers applied a range of ML approaches, such as naïve Bayes classification, RF, LR, SVM, and CART. After examining the results, it was discovered that the RF algorithm had a spectacular performance, diagnosing PCOS with 96% accuracy on the tested dataset.

Silva et al. [12] demonstrated the BorutaShap approach in their work and then proceeded to train an RF model in an orderly way. There were 72 PCOS patients, and on the other hand, there were 73 healthy women included in the dataset. There were 58 features that were graded according to their significance and relevance. At the end of the day, the model was adept at achieving an accuracy of 86%.

Mehr and Polat [13] evaluated the diagnosis of PCOS by using different classification methods on the Kaggle PCOS data. The main point of their research was to promote the use of feature selection alongside methods like ensemble learning. They checked the performance of RF, AdaBoost (AB), Extra Trees, and MLP against the whole set of features and against output from filter, embedded, and wrapper feature-selection algorithms. The RF classifier performed best, with an accuracy of 98.89%, when it was trained on a feature set that had been chosen using an integrated approach.

Adla et al. [14] developed a model that streamlines PCOS detection by using machine learning approaches. By applying a dataset containing 39 features and data from 541 patients, they placed the features according to their significance. In an attempt to minimize the quantity of attributes, the researchers proposed a hybrid feature selection approach involving wrappers and filters. Additionally, they predicted PCOS using several kinds of machine learning models using their preferred characteristics. The maximum accuracy, 91.6%, was shown using SVM.

Khanna et al. [15] proposed a methodology for estimating PCOS in people who are fertile. They used an available dataset of 541 patients from Kerala, India. The NB, KNN, LR, Extra Trees, RF, decision tree (DT), SVM, Gradient Boost, and AB ML models were used by the authors to forecast PCOS. Moreover, they proposed multistacking ML. They utilized XAI techniques that rendered model predictions reasonable, interpretable, and reliable. According to the results, multi-stacking machine learning achieved the best accuracy, at 98%.

Dutta et al. [16] built a PCOS classifier using a Synthetic Minority Oversampling Technique (SMOTE)-based workflow and several models: LR, RF, DT, support vector machine (SVM), and KNN. They first cleaned the data by imputing missing values and scaling, then reduced dimensionality with PCA. Models were evaluated with stratified *k*-fold validation using accuracy, precision, recall, F1, receiver operating characteristic (ROC), area under the curve (AUC), and runtime. LR with SMOTE delivered the best average accuracy of 95.5%; SVM and KNN achieved the highest AUC of 95.6%; and RF was the fastest at 0.010 s, outperforming the other setups.

Researchers Zigarelli et al. [17] discussed the use of ML to design a predictive system for detecting PCOS on your own. They downloaded PCOS data from Kaggle that contained 44 features and details from 541 patients at 10 institutions in Kerala. To check the model's results, the researchers applied *K*-fold CV. The system had higher accuracy (up to 92.5%) with non-invasive clinical indicators than invasive procedures (82.5%).

Denny et al. [18] developed a method to detect and predict PCOS early using minimal, although efficient, clinical and metabolic clues. The authors performed Principal Component Analysis (PCA) to reduce the number of attributes. PCOS prediction was accomplished using LR, NB, KNN, RF, and SVM, utilizing specific features. The study found that RF had the best accuracy of 89.02%.

Bhardwaj and Tiwari [19] provided a comprehensive analysis of the many clinical traits and how each influences a patient's risk of developing PCOS. They employed a Kaggle PCOS dataset, which was open-source and had 541 people's medical records along with 44 characteristics that had the goal label of PCOS diagnosis. The multicentric dataset has been obtained from eleven hospitals in Kerala. The architecture consists of the Pearson correlation approach for feature selection and ML classifiers. The most significant traits were determined to be hair growth, prolactin levels, the mean size of both right and left follicles, and the number of left follicles. Using the MLP with an SVM radial basis function kernel, a 93% accuracy rate was attained.

Tiwari et al. [20] used a clinical dataset to develop a model for detecting PCOS by Kottarathil that was made visible on its Kaggle repository. For selecting a particular group of features from the database, researchers used correlation feature selection methods. They identified the most suitable approach based on correlation thresholds by using a variety of machine learning models, among them XGB, Linear Discriminant Analysis (LDA), SVM, LR, RF, GB, AB, DT, KNN, QDA, and CB. Having an accuracy of 93.25%, the RF method beat other recognized machine learning algorithms, according to the experiments.

Bharati et al. [21] concentrated on the data-based diagnosis of female PCOS. They explored the use of a publicly accessible data set containing 541 women, 177 of whom had been

diagnosed with PCOS. A hybrid RFLR model, LR, RF, Gradient Boosting (GB), and a univariate feature selection strategy were all used by the authors. To train and evaluate the models, they split the dataset using holdout and CV techniques. The findings showed that the RFLR model with UFS had the best accuracy, 91.01%.

Rahman et al. [22] used machine learning methods in their proposal to create a web-based PCOS prediction system. Forty-three features from 541 Kaggle records were preprocessed and then combined into 42 columns for the study. Twelve major characteristics were determined using mutual information. They considered 13 classifiers, including LR, RF, AB, SVM, and DT. With Mutual Information, the outcomes showed that AB and RF had the highest accuracy, at 94%. The latest technique was demonstrated to be more accurate than earlier studies. For enhanced patient outcomes and early detection, researchers integrated these models into an intuitive Django-based internet application for real-time PCOS prediction.

A comprehensive literature review revealed that several ML classifiers, including RF, LR, CNN, SVM, and K-Means, showed inconsistent results in their application to PCOS. The variability in results presents an intriguing situation, as each method generates unique findings. The study utilizes a straightforward machine learning method focused on feature extraction, demonstrating enhanced accuracy in comparison to earlier research efforts. Table 1 shows a succinct overview of the relevant studies in this field.

3 | Materials and Methods

The entire procedural workflow of the proposed framework is illustrated in Figure 1. Initially, the dataset underwent basic preprocessing, including handling missing values and encoding categorical variables into numerical formats. In order to rigorously avoid data leakage and obtain an unbiased assessment, the dataset was then divided into an 80% training set and a 20% hold-out testing set prior to the processing using labels. Next, the SMOTE resampling method was applied to the training set to correct class imbalance. Similarly, the two-stage feature selection process was executed solely on the training set to identify the most critical predictive features. This refined training set was then used as input to train several machine learning models and ensemble voting classifiers. Their predictive performance was then assessed on the 20% test set using classification metrics and confusion matrices to determine the best-performing model architecture. Lastly, to improve the interpretability and explainability of the model for clinical purposes, XAI methods such as SHAP and LIME were incorporated. SHAP was used to calculate global feature importance, whereas LIME provided a local approximation for individual predictions. These methods provide a holistic view of the impact of physiological features on the model's reasoning.

3.1 | Dataset Description

Using a publicly accessible dataset of PCOS patients from "Kaggle," a dataset including the symptoms and PCOS diagnostic results of patients has been used as the training data for supervised machine learning models [23]. While being used for training, the dataset was carefully analyzed to increase comprehension. Based on the initial evaluation of the PCOS records, there are 541 items in the dataset pertaining to female patients, with 44 columns including various types of clinical data relevant to the PCOS disease. An evaluation result for PCOS is shown in one of the columns titled "PCOS(Y/N)," where 1 and 0 indicate whether the patient has the disorder or not. In this study, the target variable for the training has been determined to be this feature column. In the whole dataset, a score of 0 denotes a "negative" or "no" reaction, while a score of 1 denotes a "positive" or "yes" response. The information includes a variety of variables, including blood group, pulse rate, weight, height, number of abortions, cycle length (in days), pregnancy status (y/n), hair loss (y/n), acne (y/n), and more. After examining the data in this column, it was found that there is an unequal distribution of favorable and unfavorable results, with 364 entries labeled as "NO PCOS" and 177 items labeled as "PCOS." This study was carried out using patient de-identified data on PCOS that was publicly available on the Kaggle platform. No new experiments on humans or animals were done. Because only anonymized public data were analyzed in this study, formal ethics approval and individual consent were not required. The dataset is extensively used in the field of predicting PCOS and has been used as a basis for several peer-reviewed studies and academic analyses.

3.2 | Dataset Preprocessing

Initially, the null values were controlled for the purpose of preparing the data. Features having a high proportion of null values or values that are missing have been cleared from the dataset since they do not provide any useful information. For instance, since the dataset contains 539 null items, the characteristic "Unnamed" has been eliminated. Also removed from the dataset are the unnecessary columns "Sl. No." and "Patient File No." which merely provide the patient's file number and serial number, respectively, which may be ignored for further study. Some of the features' null values have been adjusted to more relevant values; for instance, "Marriage Status (Yrs)" and "Fast Food (Y/N)" both have a few null values that have been converted to the column's most common value.

Data normalization is the next step, in which the MinMax Scalar approach is used to alter the dataset values. This process attempts to reduce the effect of measurement unit variation across different features and reduce attribute bias related to sensitivity. The MinMax scaler rescales the feature range values from 0 to 1 using Equation (1). In Equation (1), x_{scaled} is the rescaled value derived from the original value x, where $x_{\min}$ and $x_{\max}$ are the lowest and largest values of that attribute A_i, respectively.

$$x_{\text{scaled}} = (x - x_{\min}) \div (x_{\max} - x_{\min}). \quad (1)$$

Furthermore, given that the dataset features an imbalanced emphasis on attributes—364 records of non-PCOS patients in addition to 177 records of PCOS patients—data maintenance has been implemented to guarantee a fair distribution of classes

TABLE 1 | Table of comparisons for existing works.

References	Dataset	Attributes used	Method	Limitations
Hassan et al. [11]	541 patient dataset from Kaggle	42	RF	Limited algorithmic evaluation (only five traditional ML models)
Silva et al. [12]	72 PCOS patients and 73 healthy women	14	BorutaShap, RF	Very small dataset
Mehr and Polat [13]	541 patient dataset from Kaggle	28	RF	Raises suspicion of overfitting or data leakage
Adla et al. [14]	541 patient dataset from Kaggle	10	Linear SVM	High precision (≈93.7%) but relatively low recall (≈80.6%), indicating imbalanced sensitivity
Khanna et al. [15]	Dataset includes 541 Indian patients from Kerala	10	Multi-stacking ML	High-performance metrics may indicate overfitting
Dutta et al. [16]	541 patient dataset from Kaggle	12	LR	Exact post-PCA feature list not fully enumerated; reliance on SMOTE may shift data distribution
Zigarelli et al. [17]	The dataset comprises 541 Kerala-based Indian patients	10	CB	Patient model: 81%–82.5%, provider model: 87.5%–90.1%, showing need for better patient-side accuracy
Denny et al. [18]	541 patient data from clinical examinations	8	RF	Comparatively low accuracy on this dataset
Bhardwaj and Tiwari [19]	Dataset of 541 patients from 10 Distinct Keralan	5	MLP, SVM, RBF	Very few number of features used
Tiwari et al. [20]	A Kottarathil clinical dataset	10	RF	Did not use any XAI
Bharati et al. [21]	541 patient dataset	10	RFLR	Missing F1-score, specificity, and confusion matrices limit performance evaluation
Rahman et al. [22]	Kaggle PCOS dataset (541 records, 42 features)	12	MI-RF, MI-AB	Did not use any XAI

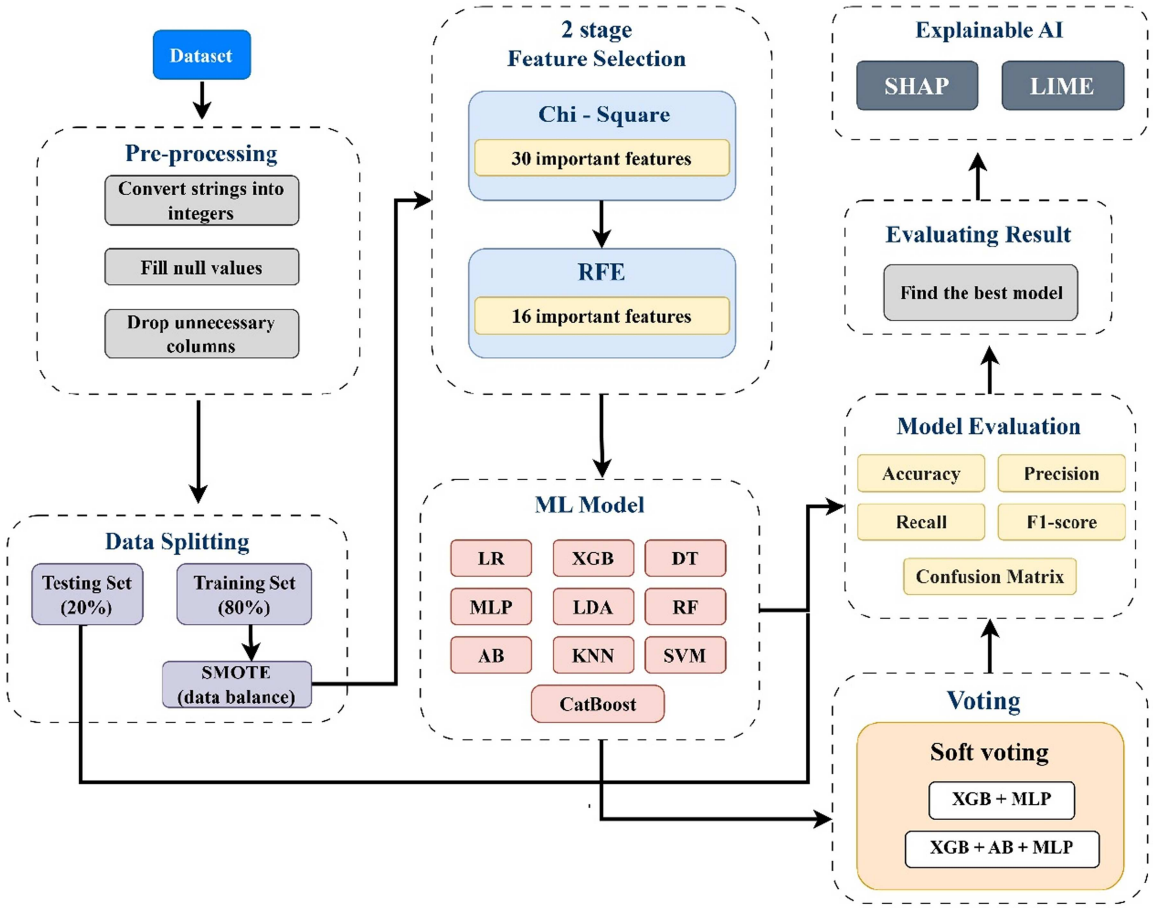


FIGURE 1 | The visual representation of the proposed method.

TABLE 2 | Dataset distribution before and after oversampling.

PCOS 541 patients' dataset					
Before oversampling			**After oversampling**		
Training	**Testing**	**Total**	**Training**	**Testing**	**Total**
432	109	541	582	109	691

for training purposes. Following initial scaling and missing-value imputation, the entire dataset was strictly partitioned into training (80%) and testing (20%) subsets prior to any label-dependent processing to prevent data leakage. To address the class imbalance without biasing the evaluation, the "Synthetic Minority Oversampling Technique (SMOTE)" was applied exclusively to the 80% training subset. Equation (2) presents the mathematical formula applied in the SMOTE method, where x_{sample} represents the sample obtained from the minority class value x, and x_{random} is a value randomly chosen from the nearest neighbors of x, constrained by $0 \leq \eta < 1$.

$$x_{\text{sample}} = x + \eta(x_{\text{random}} - x). \tag{2}$$

The dataset utilized in this study consists of 541 medical records, each detailing specific clinical symptoms and diagnostic features related to PCOS. As previously stated, the data was strictly split, allocating 80% for training models and 20% for testing. To properly address class imbalance without risking data leakage, synthetic samples were subsequently generated using the SMOTE technique exclusively within the training set. Table 2 presents the distribution of samples across the training and testing partitions before and after the application of SMOTE, clearly demonstrating that the hold-out testing set remained entirely unaltered.

3.3 | Feature Selection

To increase the prediction ability and accuracy of machine learning algorithms, feature selection is an effective method to recognize the most vital features and to abandon irrelevant ones. To provide the most accurate data description, the approach requires investigating the best selection or subsets of features. After preprocessing, the dataset in this study has 40 features, which might reduce the classifier's accuracy if all of the less important ones are included. In this context, the features have been meticulously prioritized and selected through

TABLE 3 | Feature selection breakdown: From Chi-Square filtering to wrapper model refinement.

Stage 1	Stage 2											
	Inclusion of feature in 2nd stage											
	Selected in top 20 by wrapper model				Selected in top 16 by wrapper model				Selected in top 12 by wrapper model			
Preselected 30 features by χ^2	CB	XGB	LR	RF	CB	XGB	LR	RF	CB	XGB	LR	RF
Age (years)	✓	✓	✓	✓		✓		✓			✓	✓
Weight (kg)	✓	✓		✓		✓		✓		✓		✓
Height (cm)				✓								✓
BMI				✓				✓				✓
Blood group												
RR (breaths/min)	✓	✓	✓		✓	✓				✓		
Hb (g/dL)	✓	✓	✓	✓		✓						
Cycle (R/I)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cycle length (days)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Marriage status (years)	✓	✓	✓	✓	✓		✓		✓		✓	
Pregnant (Y/N)												
No. of abortions		✓	✓				✓				✓	
II beta-HCG (mIU/mL)												
FSH (mIU/mL)	✓			✓	✓			✓	✓			
Hip (inch)				✓								
Waist (inch)	✓	✓	✓									
AMH (ng/mL)	✓		✓	✓	✓		✓	✓				
Vit D3 (ng/mL)												
Weight gain (Y/N)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Hair growth (Y/N)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Skin darkening (Y/N)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Hair loss (Y/N)												
Pimples (Y/N)	✓	✓	✓		✓		✓		✓			
Fast food (Y/N)	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	
Reg. exercise (Y/N)		✓	✓			✓				✓		
Follicle (L)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Follicle (R)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Avg. *F* size (L) (mm)	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓
Avg. *F* size (R) (mm)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
Endometrium (mm)	✓	✓	✓	✓		✓	✓	✓				✓

the application of three distinct feature selection techniques to identify the optimal set of features from the PCOS data set. The methods will be outlined in the following sections:

- **RFE Technique:** Recursive Feature Elimination, or RFE, is a highly effective wrapper-type approach used in this study for the removal of features from a training sample throughout the feature selection process. Using this strategy, the traits that add less to the classification are gradually removed from the bottom of the ranking [24]. This method fundamentally involves a recursive process that utilizes multiple ML techniques as its core, arranged within the RFE framework. As a result, feature importances are determined during each iteration, leading to the elimination of the least relevant features in order to select the prioritized ones [25]. In this case, the RFE method from the scikit-learn Python ML library's RFE class has been used for implementation. Table 3 displays the three set features selected using the RFE feature selection technique.
- **Chi-Square Technique:** In machine learning, the Chi-Square feature selection method is one of the most popular and useful feature selection methods. Prioritizing features by examining their relationships with one another, it then runs a numerical test to identify divergence from the predicted distribution in cases when the feature event is unrelated to the class value [26]. The Chi-Square feature selection formula is shown in Equation (3). According to the equation, each observed value shows how many observations are joined by a certain characteristic, and the expected value gives the predicted outcome. Here, the highest χ^2 scores are used to determine which attributes are

prioritized. To implement the Chi-Square numeric test, the Python function "SelectKBest" was used. This method uses $k = n$, where k is the number of features that the algorithm would choose. Then, n features were chosen from the dataset's 40 features based on the highest scores. Table 3 illustrates the features that were selected using the χ^2 feature selection approach.

$$\chi^2 = \sum_{i=1}^{n} \frac{\left(\text{Observed value}_i - \text{Expected value}_i\right)^2}{\text{Expected value}_i}. \quad (3)$$

- **Chi-Square + RFE:** By combining statistically significant outcomes with model-based feature evaluation, Chi-Square, tackled with RFE, provides a significant advantage. Furthermore, to be statistically significant, the features obtained by this hybrid method also make a significant boost to the machine learning model's accuracy in predicting [27]. The chi-Square test minimizes the feature space and calculates cost by discovering features as a preselection filter that have an important statistical correlation with the target variable. After that, RFE frequently eliminates the least significant features from this subset based on the way they predict the model. By emphasizing the most significant and relevant subset of features, this hybrid method increases model accuracy, reduces overfitting, and improves computing efficiency. Table 4 illustrates the combined characteristics of the features chosen using Chi-Square and RFE.

3.3.1 | *Feature Selection Results Using Two-Stage Feature Selection Method*

For the classification of PCOS, we employed a systematic two-stage feature selection process to identify the most relevant features for our proposed methodology. This was made at the compromise between the quality of the model and computing economy aimed at avoiding the blow of dimensionality while keeping a fine level of predictive power. The engineering features were selected to improve interpretability, lower the chance of overfitting, and enhance the model's capacity to accurately detect PCOS. This way, though, we were able to find something quite solid that would give reliable results when deployed.

TABLE 4 | Final selected features using RFE.

Sl. No.	CB 16
1	Follicle (R)
2	Follicle (L)
3	Hair growth (Y/N)
4	Cycle (R/I)
5	Weight gain (Y/N)
6	Skin darkening (Y/N)
7	Cycle length (days)
8	Pimples(Y/N)
9	Avg. *F* size (L) (mm)
10	Marriage status (years)
11	Avg. *F* size (R) (mm)
12	RR (breaths/min)
13	Fast food (Y/N)
14	Weight (kg)
15	FSH (mIU/mL)
16	AMH (ng/mL)

Stage 1 of Feature Selection—41 features were initially extracted in Stage 1, and their statistical significance with respect to the target class was evaluated using the Chi-square test. Chi-square test is a statistical test commonly used to find out whether a feature is closely related to the target, as well as test independence between categorical variables. By analyzing the *p*-values produced by the test, attributes of high importance were selected, while those with weak correlations or no impact on the model's predictive performance were eliminated. This technique reduced the feature set to 30, retaining only the most relevant features for further processing. The Chi-square test played a pivotal role in confirming that the selected features had strong correlations with the target variable, laying the groundwork for the subsequent stages of the feature engineering process.

Stage 2 of Feature Selection—In the second stage, RFE was implemented to select informative features iteratively. To evaluate the performance with fundamental stability, four independent RFE methods were employed: CB, LR, XGB, and RF. These methods are used for feature selection on different subsets, such as 20, 16, and 12 features. This led to 12 differently generated feature sets (4 techniques × 3 subset sizes) and potentially a different viewpoint about the most important features. This was realized as best as possible for the prediction task, using a trade-off of approaches with varying subset sizes.

In conclusion, following our two-stage feature selection process, we identified 16 prominent features as predictive candidates of PCOS. As highlighted in Figure 2 and Table 4, and further supported by the statistical summaries presented in Table 5, these distilled features demonstrate significant discriminative power between PCOS and non-PCOS groups. In this case, β is the estimated effect size (regression coefficient on the log-odds scale); β and 0 will increase the odds of PCOS, and 0 – negatively. *Y*-correlation represents the correlation between each feature and the binary outcome (PCOS = 1), with the farther the value is from +1 or −1 indicating stronger positive or negative correlation, and the closer to 0 the value, the stronger the null hypothesis is not associated with this feature. p represents the *p*-value of the association between the feature and the outcome, where small values (e.g., $p < 0.05$) show more evidence against the lack of association between the null hypothesis and that feature. Y1-mean is the average of the attribute in PCOS cases ($Y = 1$) in binary cases, or the percentage whose value is 1 in the PCOS group, and Y0-mean is the percentage whose value is 0 in the control group ($Y = 0$). All mean refers to the average sample mean of the characteristic of all the subjects, and All Std refers to the average standard deviation. By utilizing Chi-square analysis and RFE with multiple models, we ensured that the selected features remained both stable and explainable, despite the relatively small dataset size. This optimized set of statistically validated features provides a robust foundation for our

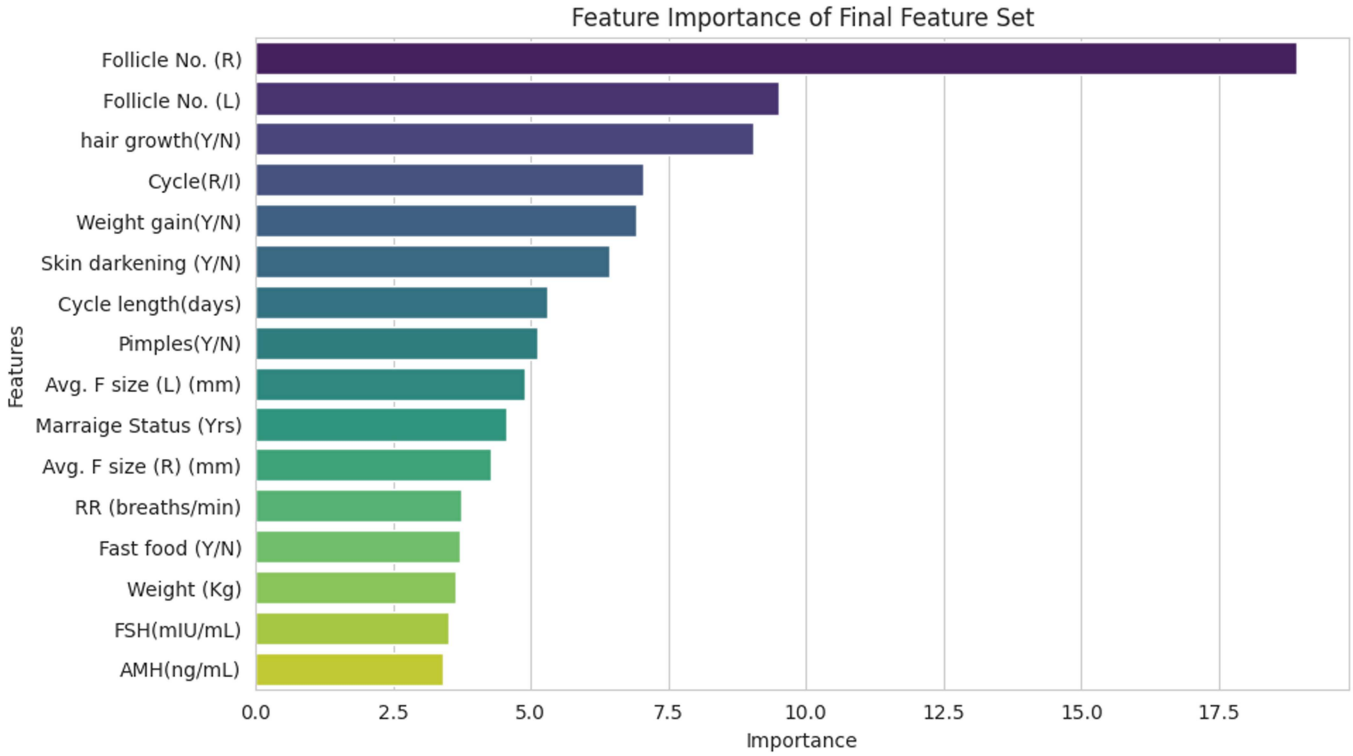


FIGURE 2 | Selected important features graph using RFE.

TABLE 5 | Statistical summary of common features.

Feature	β	β-CI-lower	β-CI-upper	*Y*-correlation	*p*	Y1-mean	Y0-mean	All mean	All Std
RR (breaths/min)	0.123	−0.159	0.405	0.037	0.385	0.278	0.267	0.270	0.141
Cycle (R/I)	0.627	0.506	0.748	0.402	0.000	0.360	0.103	0.187	0.301
Cycle length (days)	−0.674	−0.989	−0.360	−0.178	0.000	0.380	0.427	0.412	0.124
Marriage status (years)	−0.336	−0.583	−0.090	−0.115	0.007	0.230	0.269	0.256	0.160
FSH (mIU/mL)	−0.331	−1.256	0.593	−0.030	0.313	0.001	0.004	0.003	0.043
AMH (ng/mL)	0.107	−0.030	0.245	0.066	0.141	0.530	0.490	0.503	0.288
Weight gain (Y/N)	0.427	0.353	0.500	0.441	0.000	0.684	0.228	0.377	0.485
Hair growth (Y/N)	0.489	0.410	0.568	0.465	0.000	0.571	0.129	0.274	0.446
Skin darkening (Y/N)	0.484	0.408	0.560	0.476	0.000	0.621	0.154	0.307	0.462
Pimples (Y/N)	0.268	0.192	0.345	0.286	0.000	0.695	0.390	0.490	0.500
Fast food (Y/N)	0.353	0.280	0.427	0.376	0.000	0.785	0.385	0.516	0.500
Follicle (L)	1.474	1.309	1.639	0.603	0.000	0.445	0.198	0.279	0.192
Follicle (R)	1.372	1.236	1.509	0.648	0.000	0.538	0.232	0.332	0.222
Avg. *F* size (L) (mm)	0.420	0.155	0.685	0.133	0.001	0.654	0.612	0.626	0.149
Avg. *F* size (R) (mm)	0.332	0.046	0.618	0.098	0.019	0.663	0.634	0.644	0.138
Endometrium (mm)	0.416	0.088	0.745	0.107	0.009	0.489	0.462	0.471	0.120

classification model, achieving a balance between dimensionality reduction, predictive accuracy, and generalizability.

3.4 | Machine Learning Classifier

Machine learning models help systems predict results, using data analysis to make forecasts in many different domains more accurate [28]. In this study, performance has been evaluated using 11 classifiers, which are LR, DT, RF, GB, AB, XGB, SVMs, KNN, Naive Bayes, MLP, and Bagging. The best model with the highest accuracy among all classifiers is then chosen via voting.

3.4.1 | *LR*

LR, a commonly used supervised learning algorithm, is used to work out the chance of a dependent variable given different independent or factor data [29]. Many people use LR as a

statistical technique. It gives the possibility that an instance represents a certain class. The basis of LR is the logistic function, sometimes known as the sigmoid function.

3.4.2 | DT

A much-used technique in supervised learning for classification and regression is known as DT. By dividing the feature space into many smaller parts, it can judge data points using each feature's value. Every leaf node shows the prediction or class result, and every internal node is a choice based on a specific feature [30]. Equation (5) represents DT classifier and A'' gives the accurate decision.

$$A''(a_i) = [D_0(a_i) - D(a_i)] + K[\in_0 (a_i) - \in (a_i)] + \sum_{j=1}^{b_i} x_i + {}_j y_0(a_{i+j}). \quad (5)$$

Considering a dataset $D = \{(a_1, b_1),(a_2, b_2), ..., (a_n, b_n)\}$, where b_i is the associated class label, and a_i is the feature vector. The DT learns a function $f: x \rightarrow y$, where x is the space of features and y is a collection of class labels [31].

3.4.3 | RF

RF is formed of several sets of DT. By measuring information gain, it identifies the most beneficial collection of features. The approach classifies an instance based on a majority judgment among numerous DTs. Combining the average prediction for regression or the median of categories for classification predictions gleaned from every tree yields the final prediction [32]. In simpler terms, the node probability may be determined by splitting the entire amount of samples gathered by the total amount of samples that reach the node. An attribute's relevance increases with its value.

3.4.4 | LDA

LDA is a supervised learning technique for dimensionality reduction and classification. When moving high-dimensional data into a lower-dimensional space, it improves class separability. Each class is assumed by LDA to have the same covariance matrices and a Gaussian distribution. It allocates new instances to the class with the greatest posterior probability after computing a linear decision boundary using Bayes' theorem. It improves computing efficiency while preserving classification accuracy, making it especially useful for high-dimensional datasets. Speech recognition tasks, medical diagnostics, and facial recognition all make heavy use of LDA [33].

3.4.5 | AB

In order to enhance its accuracy, the AB classifier, which acts as a meta-estimator, combines many classifiers after first training a classifier utilizing the original dataset. Its basic principle is to guarantee accurate predictions, especially for rare cases, by fine-tuning data training and modifying classifier weights at each stage [34]. AB also gives classifiers weights depending on their accuracy, giving greater weight to those that do better in each iteration.

TABLE 6 | Software libraries and versions used in the analysis.

Library	Version	Library	Version
pandas	2.2.2	LightGBM	4.6.0
numpy	2.0.2	XGBoost	3.1.1
scikit-learn	1.6.1	SHAP	0.49.1
imbalanced-learn	0.14.0	SciPy	1.16.3
matplotlib	3.10.0	statsmodels	0.14.5
seaborn	0.13.2	LIME	0.2.0.1

TABLE 7 | System environment configuration.

Resources	Details
CPU	Intel Core i3-6006 CPU @ 2.00 GHz
GPU	Intel HD Graphics 520
RAM	4 GB
Experimental tools	Google Colab, Jupyter Notebook, Kaggle

3.4.6 | SVMs

A technique for supervised learning used in regression and classification is called an SVM. Identifying the most suitable hyperplane for splitting data points into distinct categories is the primary objective. Using kernel functions makes SVMs capable of dividing datasets that both are and are not divided linearly, even when there are many features involved [35]. The main goal for an SVM is to identify a hyperplane that gives the largest gap between the plane and the closest points from each class [36].

3.4.7 | KNN

The non-parametric KNN technique is one of the supervised learning strategies that discovers commonalities in the training set [37]. The KNN is a well-liked and simple supervised learning algorithm for classification and regression applications. It is a non-parametric, instance-based method that classifies new instances based on their similarity with those in the training dataset [38].

3.4.8 | XGB

XGB is a high-performance implementation of only the gradient boosted DT. It is widely used for classification and regression due to the fast training rate and high predictive accuracy [39]. XGB constructs an ensemble in a stagewise way, by constructing a large number of shallow trees sequentially. Each new tree is fitted to the residual mistakes of the previous model, and progressively rectifies previous errors. One of the main characteristics is that it has a regularized objective that balances the goal of fit and simplicity so that overfitting is avoided. The objective is a combination of a loss term (to measure the

training error) and a regularization term (to penalize excessive tree complexity), resulting in models that generalize well.

3.4.9 | MLP

The MLP is a feedforward neural network that uses many connected layers. It considers the node values, the activation function, the Sigmoid function, and the weights of the input layer to calculate the output. Propagation in the network takes place because the input layer provides values to the hidden layer, which then moves on to the output layer. Each neuron in an MLP is fully connected to the neurons in the subsequent layer, enabling complex pattern recognition and feature extraction. Depending on the problem's characteristics, MLP may make use of multiple activation functions, including ReLU, Tanh, or Softmax. It also uses gradient descent and backpropagation to minimize the loss function and modify the weights, thereby making it an effective model for supervised learning tasks like regression and classification [40].

3.4.10 | CB

CB is an outstanding performance gradient boosting technique that requires very little planning and is intended to handle category information effectively. CB is among the most commonly utilized classifiers. In basic terms, this method uses gradient boosting. Yandex scientists and engineers created CB [41]. It utilizes symmetric trees for quick training and inference while ensuring high accuracy, and ordered boosting to reduce target leakage. In contrast to conventional boosting techniques, CB is quite effective with structured data because it uses random variations and Bayesian bootstrapping to minimize overfitting. CB performs extremely well in both classification and regression problems, and has applications in self-driving cars,

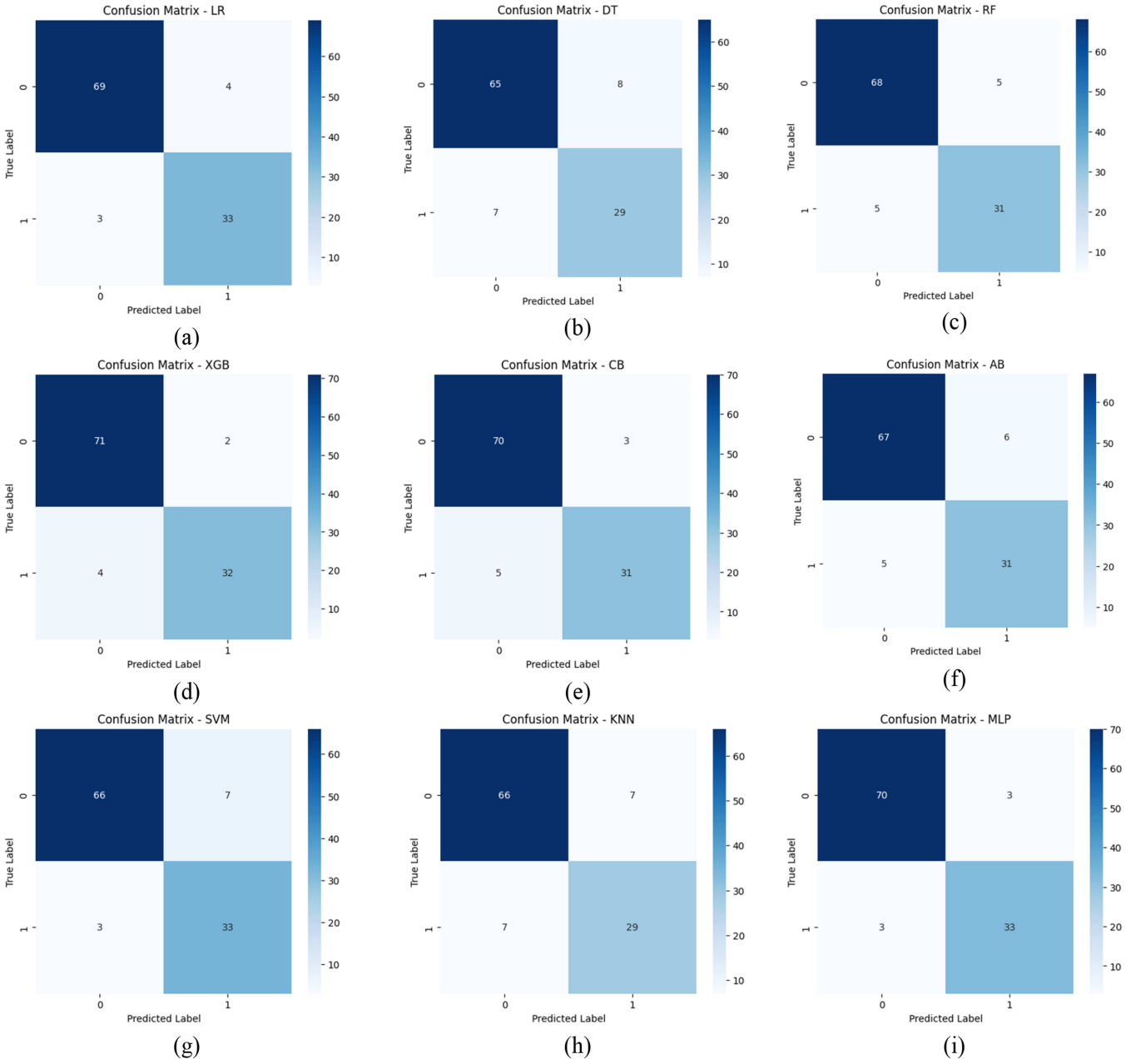


FIGURE 3 | Confusion matrix for traditional ML model using final feature set (a–i).

weather forecasting, and recommendation systems. It is a top option for machine learning practitioners looking for accurate and flexible models for prediction because of its GPU acceleration and efficient handling of big datasets.

3.5 | Statistical Analysis

The pre-specified primary analysis was binary classification of PCOS (yes/no) using the supervised algorithms described in Section 3.4: LR, DT, RF, XGB, CB, LGBM, AB, SVM, KNN, and MLP and the study also use staking ensemble. An 80/20 train–test split was used in this study. To further rigorously evaluate model generalizability and mitigate the risk of overfitting inherent to a single data split, a 10-fold stratified CV process was additionally implemented across all evaluated models. With the two-stage feature selection (Chi-Square → RFE), all preprocessing, like missing-data handling, Min–Max scaling, was performed in both the train and test sets, but the class-imbalance handling, SMOTE, was performed on the training data only to prevent leakage. Model performance on the held-out test set was summarized using accuracy, precision, recall, sensitivity, F1-score, and AUC, with confusion matrices and ROC curves reported. The results of feature-outcome associations were analyzed using univariate LR, and we present the regression coefficient (β), its 95% confidence interval (Wald), correlation with the binary outcome, and two-sided p-values with 0.05 alpha. These were the descriptive screening tests, which were not adjusted for multiplicity, and thus, the interpretation of results should be undertaken with caution. Models were not compared using any inferential test; performance of models was measured considering the held-out test set

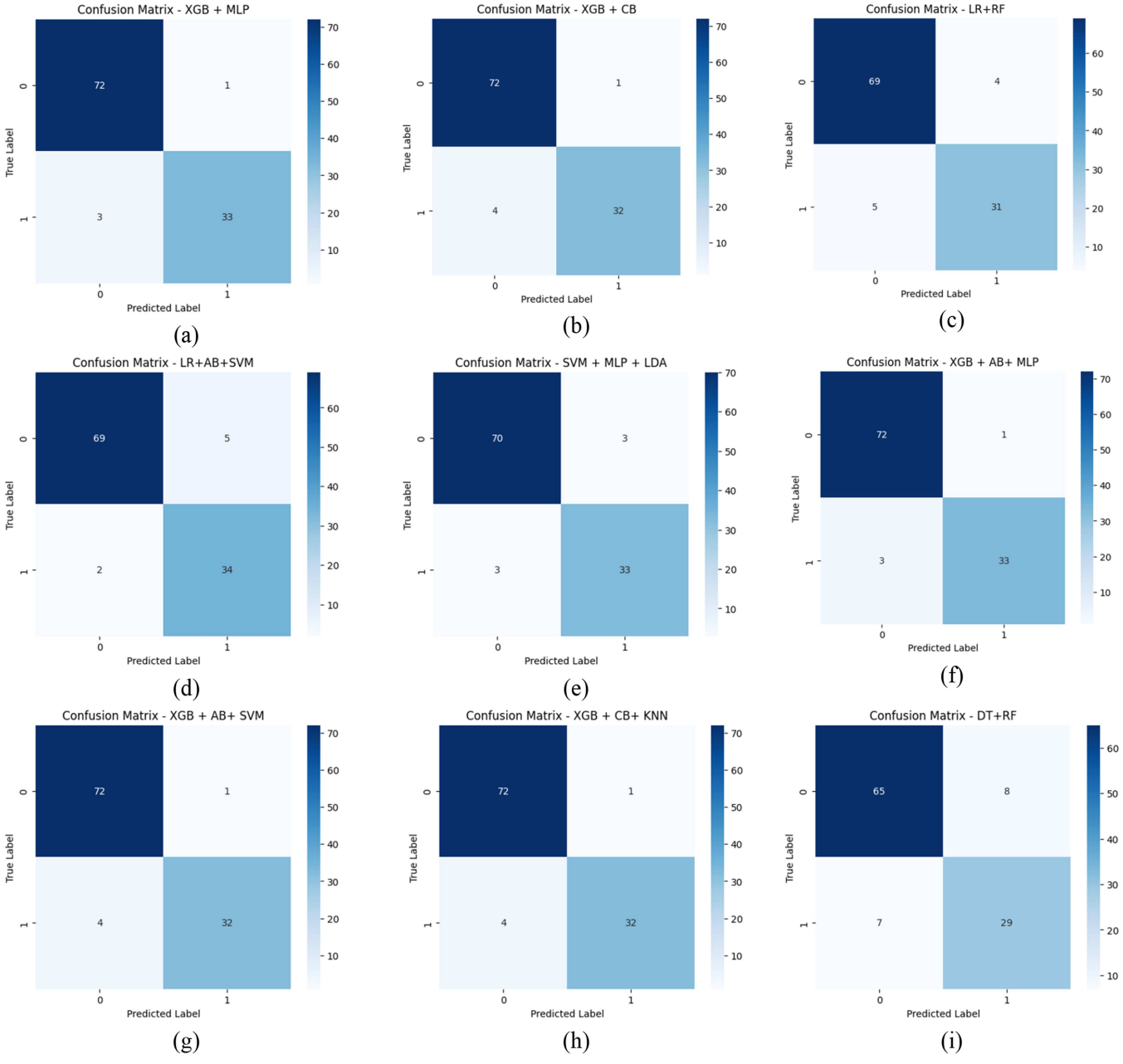


FIGURE 4 | Confusion matrix for some voting ensemble model using final feature set (a–i).

performance based on the following metrics: accuracy, precision, recall, F1-score, and the Area Under the ROC, and confusion matrices were provided. Analyses were conducted in Python language using Google Colab. The complete list of software packages and exact versions is summarized in Table 6. Reporting follows SAMPL recommendations, and standard statistical references are cited. Abbreviations are defined at first use. Also, this study adheres to the TRIPOD guidelines for reporting prediction model development and validation.

4 | Result and Discussion

The proposed method's results and discussions will be described in this section. LR, DT, RF, CB, AB, SVM, KNN, MLP, and LDA are some of the machine learning models whose outcomes for classification are shown here. Using the confusion matrix, we tested each model separately.

4.1 | System Specification

To conduct this study, a few resources were needed. This proposed framework was developed using the resources indicated in Table 7.

4.2 | Confusion Matrix

To examine the performance of the prediction analysis, Accuracy, Precision, Sensitivity (recall), and F1 score are used to check how well many machine learning models work with different sets of features [42]. For these performance metrics, the expected and actual results are compared, dividing the training set into four categories: a False Positive (FP) occurs when the real value is negative even though the predicted outcome was positive; a True Negative (TN) occurs if both the original and calculated results are negative; a True Positive (TP) happens when both are positive; and a final case of False Negative (FN) happens if the outcome is positive when the prediction was negative.

The performance of bidirectional ensemble models and standard machine learning models applied to the proposed feature set in classification is shown as confusion matrices in Figures 3a–i and 4a–i. Although XGB and MLP have a high precision, RF, DT, and LR have fair precision. The ensemble models, especially XGB + MLP and XGB + CB, are highly capable of reducing misclassifying errors, leading to improved classification accuracy. These results illustrate how well the proposed feature selection method enhances the predictive performance. Additionally, multiple-classifier models such as XGB + AB + SVM—show better generalization and stability. An

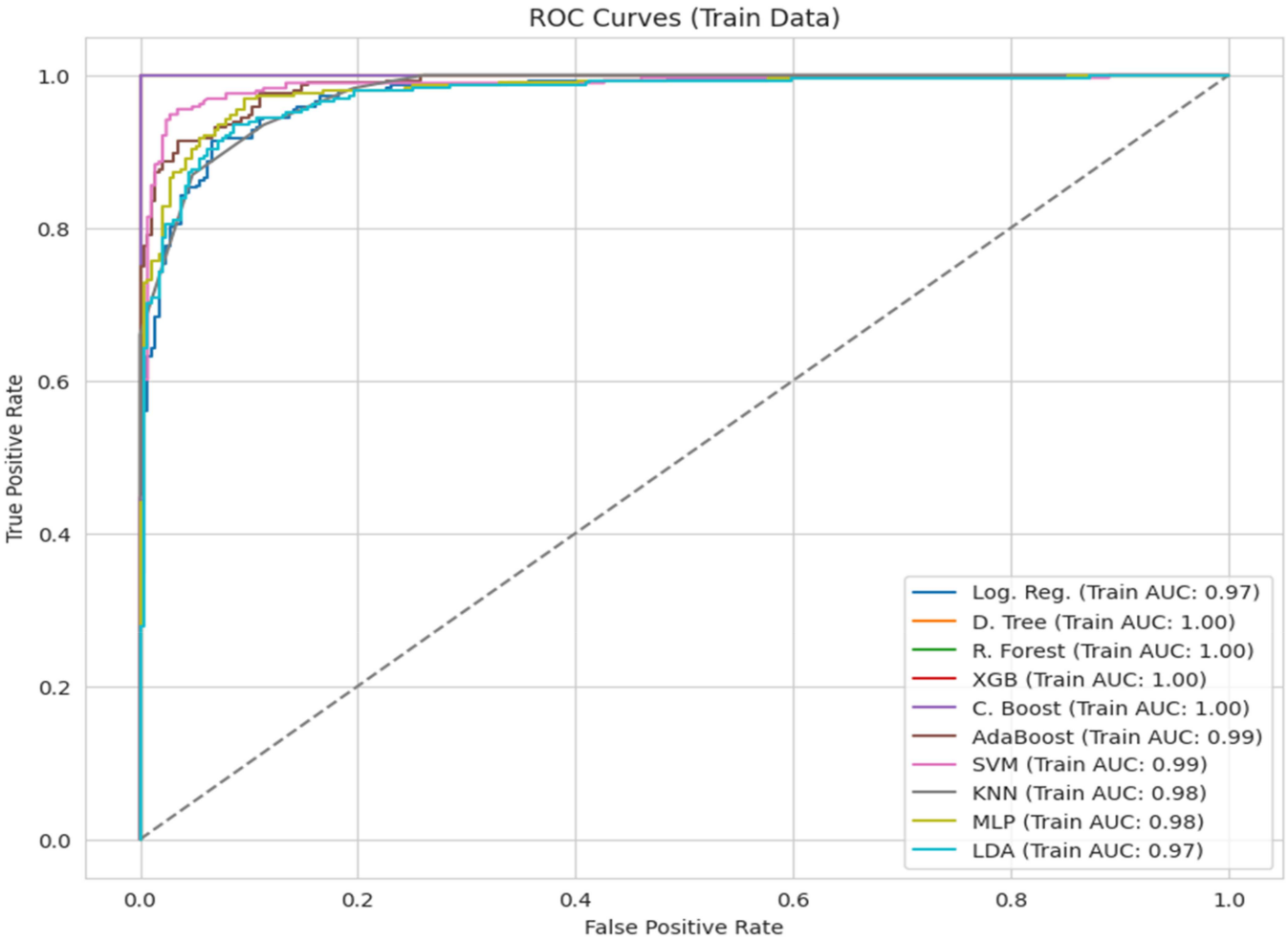


FIGURE 5 | AUC and ROC curves for the traditional ML model (train data).

extra evidence of the robustness of the selected features to PCOS classification is provided by the lower rates of false positives and false negatives, registered by these ensemble models. Moreover, ensemble methods have outperformed individual classifiers across the board, indicating the capability to leverage one approach's strengths while mitigating its vulnerabilities.

4.3 | AUC and ROC

The ROC curve is one of the most important graphical tools for analyzing the classification performance of machine learning models. The model's class-separation capabilities may be seen visually by contrasting the True Positive Rate (Sensitivity) versus the False Positive Rate (1-Specificity) under various threshold conditions. A model's usefulness can also be measured numerically using the AUC; a number nearer 100% denotes a stronger classifier. AUC values below 70% imply worse classification ability, and between 90% and 100% show great performance; also, between 80% and 90% indicate acceptable performance. The ROC curves for common machine learning models on test and training data sets are illustrated in Figure 5 and 6. An AUC equal to 100% on the training set indicates that the RF and DT classifiers proved that they categorized perfectly within the dataset. Moreover, SVM, MLP, and LDA have retained AUC scores in a range between 97% and 99%, demonstrating their significant predictive ability as well. However, only a minor decrease was observed when tested on unseen data. The AUC values for both the SVM, XGB, and LR classifiers were between 95% to 99%, indicating good but slightly lower generalization. The general investigation emphasizes the fact that ensemble-based models, specifically models with XGB, work better than standard machine learning models. Figures 7 and 8 show the ROC curves of the voting ensemble model built on training data and test data. Also, an AUC of 100% was already achieved from various combinations of RF + LR versus XGB + MLP or XGB + CB versus DT + RF versus KNN. Models based on the ensemble approach showed their superiority, evidenced by the performance of SVM + MLP + LDA, with AUC 98%; SVM + MLP + LDA, for example, had 96% AUC on data not seen during training (probably with minor changes due to overfitting to some extent or due to the difference in the dataset). The above results highlight the importance of employing ensemble approaches for learning in

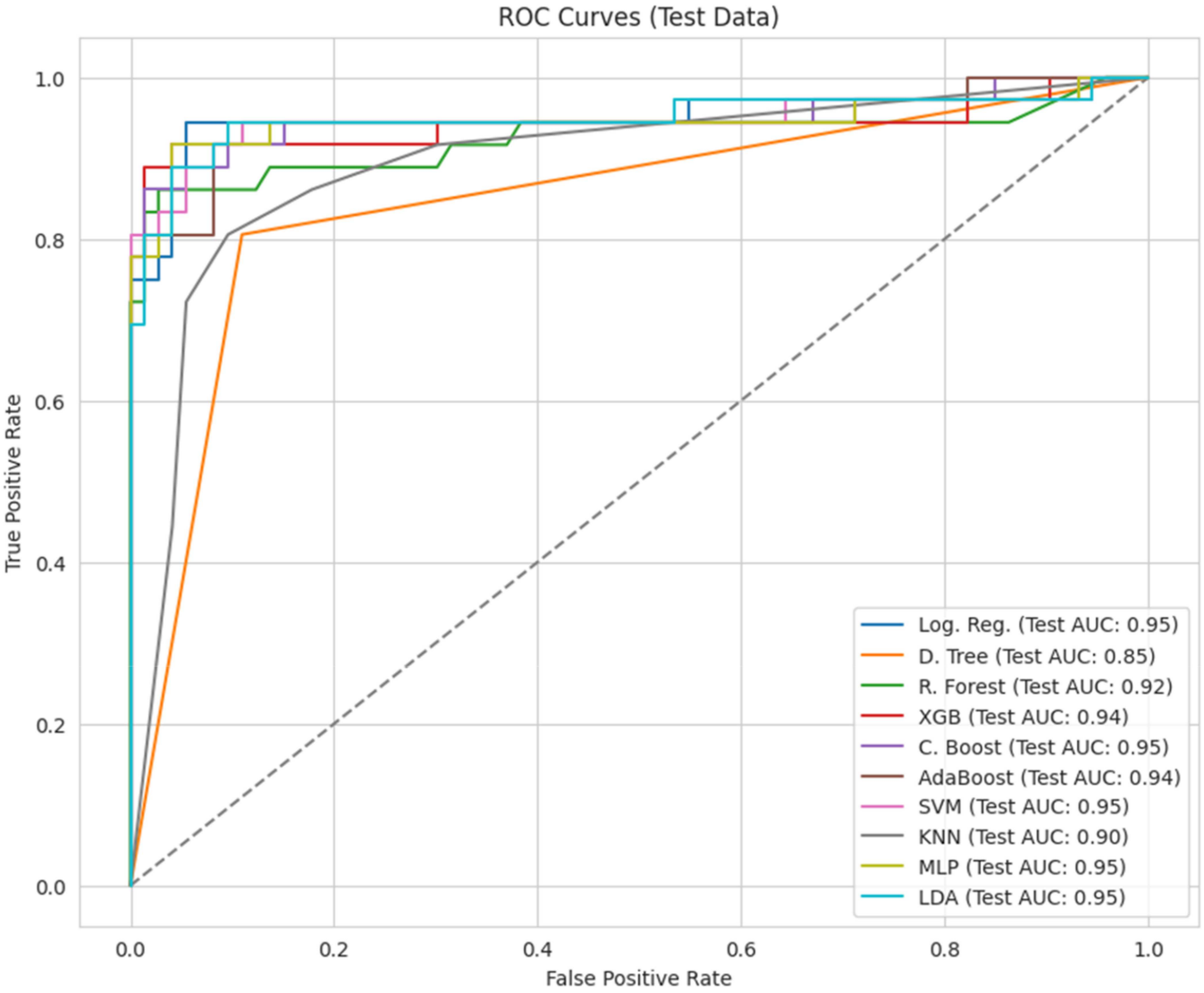


FIGURE 6 | AUC and ROC curves for the traditional ML model (test data).

order to enhance classification accuracy, especially in complex real-world applications, where improving prediction performance is crucial. Predicted ROC curves were plotted, and AUC was calculated on the test set that was held out to produce a threshold-independent measure of discrimination. This work calculated AUC confidence intervals using nonparametric bootstrap resampling of the test set and provided them together with the point estimates. In the cases of the thresholded summaries, the operating threshold was chosen within the training split by maximizing Youden J, and fixed when both the training and test splits were examined to prevent leakage. The difference between training AUC and testing AUC in Figures 5 and 6 shows that some model classes, DT and RF, fall short of their sample-size-perfect or nearly perfect training performance to significantly lower values when tested, indicating overfitting, whereas LR, SVM, MLP, and LDA are able to maintain their test AUCs near 0.95 and thus, they are better generalizers. As Figures 7 and 8 demonstrate, the ensemble combinations notably XGB + MLP are almost perfect in performance in terms of AUC on the test data, which agrees with the variance-reduction advantages of ensembling. ROC curves, which increase rapidly as one approaches the vertical axis from a screening point of view, indicate the potential to obtain high sensitivity with low false-positive rates. Since AUC is not strongly influenced by the prevalence of classes, we present AUC as total discrimination and then indicate sensitivity, specificity, and F1 to characterize workable operating points. The confidence intervals are broader due to the small size of the test set, so the external and temporal validation on independent cohorts is necessary (and indicated as a priority).

4.3.1 | *Performance Evaluation of Final Feature Set*

The classifier performance of the feature sets was evaluated primarily based on the classification accuracy and is presented in Tables 8 and 9. The results indicated a superior performance of 16 feature subsets over those with 20 and 12 features in most models, showcasing the goodness of the feature subset in terms of maintaining a fine balance between dimensionality reduction and predictive power (Figure 4a). The last step of a two-stage feature selection process we propose outputs feature sets of 16, resulting in better performance than 20- and 12-feature sets, as visually illustrated in Figures 9–12. This comparison is done using four different wrapper models in the

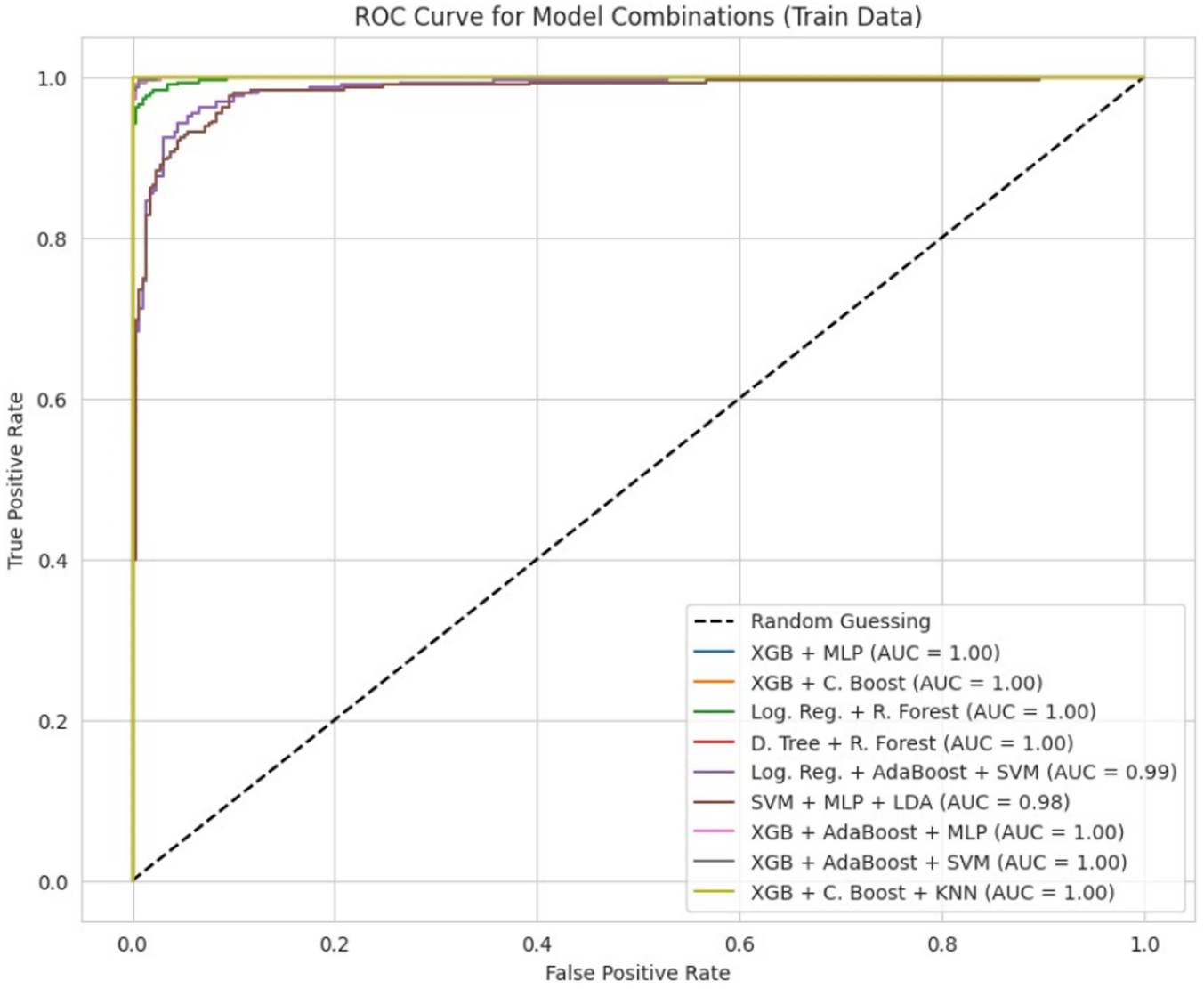


FIGURE 7 | AUC and ROC curves for the voting ensemble model for train data.

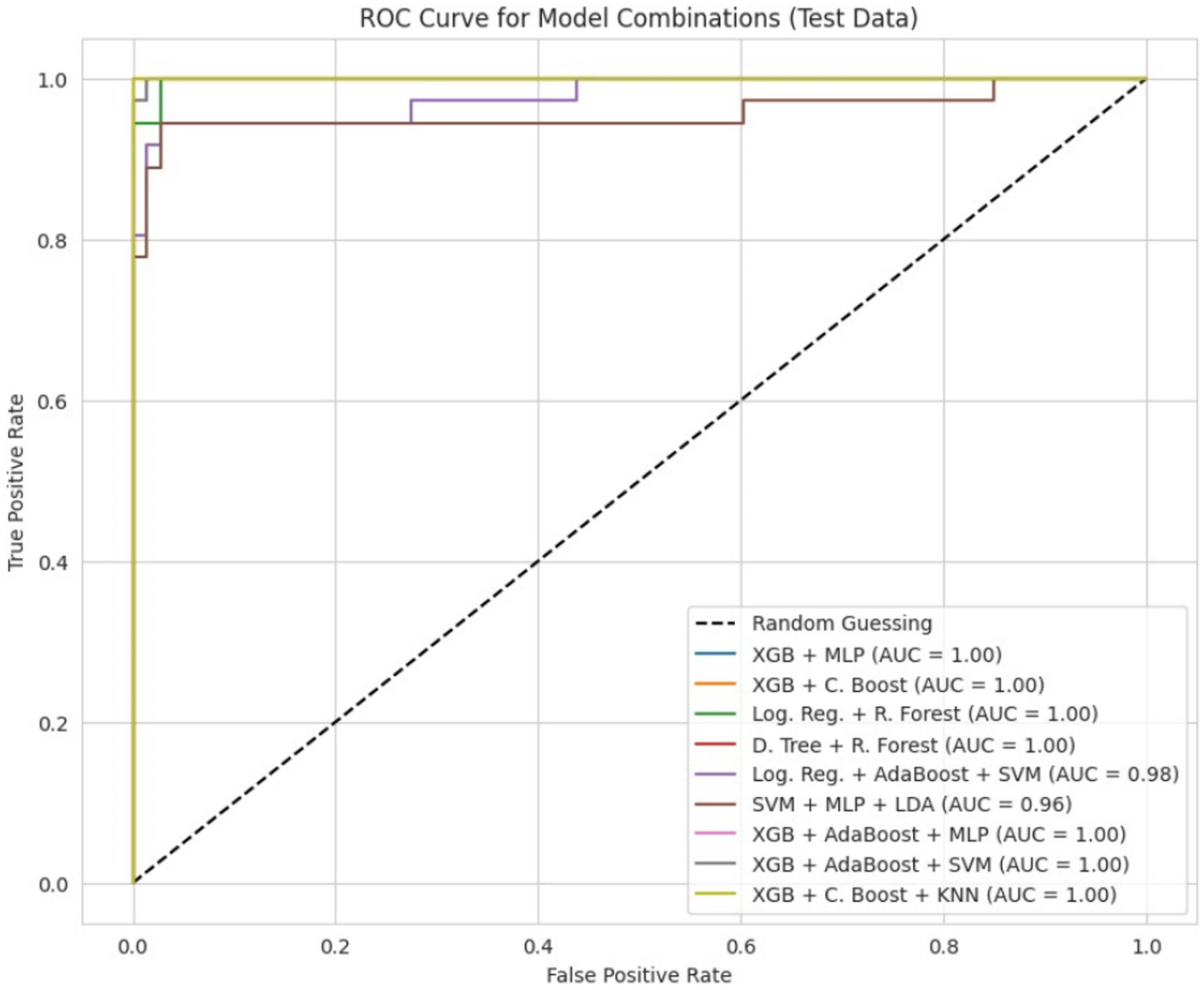


FIGURE 8 | AUC and ROC curves for the voting ensemble model for test data.

TABLE 8 | Model's accuracy of different ML models on various feature sets selected at the final stage of the two-stage feature selection.

	Wrapper model used in RFE											
	Using 20 features				Using 16 features				Using 12 features			
Model	**LR (%)**	**XGB (%)**	**RF (%)**	**CB (%)**	**LR (%)**	**XGB (%)**	**RF (%)**	**CB (%)**	**LR (%)**	**XGB (%)**	**RF (%)**	**CB (%)**
LR	94.50	93.58	94.50	94.50	95.41	93.58	94.50	93.58	94.50	93.58	92.66	91.74
DT	84.40	78.90	80.73	82.57	81.65	85.32	85.32	86.24	85.32	86.24	88.07	88.07
RF	90.83	91.74	91.74	91.74	92.66	90.83	93.58	90.83	91.74	90.83	92.66	92.66
XGB	89.91	89.91	93.58	93.58	93.58	91.74	93.58	94.50	92.66	93.58	91.74	92.66
CB	91.74	88.07	89.91	88.99	91.74	90.83	92.66	92.66	91.74	92.66	90.83	91.74
AB	90.83	88.07	88.99	88.99	89.91	88.99	89.91	88.99	89.91	89.91	88.07	89.91
SVM	93.58	90.83	91.74	91.74	91.74	91.74	93.58	90.83	90.83	90.83	92.66	91.74
KNN	83.49	86.24	87.16	85.32	87.16	87.16	87.16	87.16	83.49	87.16	88.07	85.32
MLP	93.58	92.66	93.58	93.58	94.50	93.58	92.66	94.50	92.66	91.74	92.66	90.83
LDA	91.74	90.83	92.66	92.66	91.74	92.66	93.58	92.66	92.66	90.83	92.66	90.83

TABLE 9 | Model's accuracy of stacking ensemble classifiers on different feature sets selected at the final stage of two-stage feature selection.

Model	Wrapper model used in RFE									
	Using 20 features				Using 16 features				Using 12 features			
	LR (%)	XGB (%)	RF (%)	CB (%)	LR (%)	XGB (%)	RF (%)	CB (%)	LR (%)	XGB (%)	RF (%)	CB (%)
XGB + MLP	92.66	90.83	93.58	93.58	93.58	91.74	92.66	96.33	90.83	92.66	92.66	93.58
XGB + CB	91.74	88.99	94.50	94.50	93.58	91.74	93.58	95.41	92.66	93.58	92.66	93.58
LR + RF	92.66	91.74	92.66	93.58	92.66	91.74	91.74	91.74	92.66	91.74	91.74	90.83
LR + AB + SVM	94.50	93.58	94.50	94.50	94.50	91.74	94.50	93.58	92.66	92.66	94.50	91.74
SVM + MLP + LDA	93.58	92.66	93.58	92.66	95.41	92.66	94.50	94.50	92.66	91.74	93.58	91.74
XGB + AB + MLP	92.66	90.83	93.58	93.58	93.58	91.74	92.66	96.33	90.83	92.66	91.74	92.66
XGB + AB + SVM	92.66	90.83	93.58	92.66	93.58	89.91	92.66	95.41	90.83	93.58	93.58	91.74
XGB + CB + KNN	92.66	90.83	92.66	91.74	92.66	90.83	91.74	95.41	91.74	93.58	90.83	93.58

RFE procedure. The results are clear as the 16-feature feature set outperforms the others in every single metric, strongly demonstrating the merit of our enhanced feature selection methodology. Then, with 16 chosen features using LR, LR showed the maximum accuracy of 95.41% among the other models, which was a significantly increase from its previous accuracy with 20 and 12 features (both 94.50%). Likewise, employing the 16-feature subset where CB acted as a wrapper in RFE, the grouping of XGB and MLP achieved the highest accuracy viz a viz this model with 96.33%, improving results for both of its outputs for 20 and 12 features (93.58% each).

Ensemble methods further confirmed this pattern. XGB + CB similarly achieved the best performance, achieving an accuracy score of 95.41% with a subset of the same size, while SVM + MLP + LDA reached 16 features with an accuracy score of 94.50%. The most powerful DT-based models also outperformed, achieving 86.24% accuracy on 16 features instead of 84.40% on 20 features and 88.07% on 12 features.

These results clearly demonstrate that the selected set of only 16 features represents the optimal compromise between predicting efficiency and model complexity when classifying PCOS. Using this combination of features, the proposed method achieves comparative accuracy while retaining computational economy, making it appropriate for practical uses in PCOS detection. In addition, the simplicity of the feature selection further validates its efficacy in practical clinical situations, mitigates data redundancy, increases interpretability, and provides a strong generalization across datasets.

Table 10 shows a statistical comparison of all informative selected features between classifiers LR, AB, DT, LGBM, RF, LDA, XGB, and MLP after feature selection. Reported metrics comprise β, Y-correlation, two-sided p-values, class-specific means (Y1/Y0), overall mean, and SD. Here are some features that some models exhibit strong positive associations with PCOS, whereas others show a negative association.

4.3.2 | *Final Feature Set Justification*

These findings clearly indicate that choosing the smaller selection of 16 characteristics provides an exceptional balance between having good predictive power and low complexity of the PCOS model. Notably, a maximum accuracy of 96.33% was achieved with 16 features selected by CB used in combination with XGB and MLP (Tables 8 and 9). Moreover, as the wrapper model used in RFE, at the final step of our proposed feature selection procedure, LR also performed better within the 16-feature group, yielding 95.41% classification accuracy. This optimized set of features is then utilized by the proposed technique in order to ensure high accuracy as well as computational efficiency, making it suitable for real-world applications of PCOS detection.

Table 11 displays the main results for the 16-feature set through the use of various wrapper models in RFE, but further inspection reveals additional information. Out of all wrapped models, CB produced the best accuracy values, with an average of 92.47% and 96.33% for the first two positions. Notably, CB had 94%+ accuracy in eight scenarios, while LR and RF only

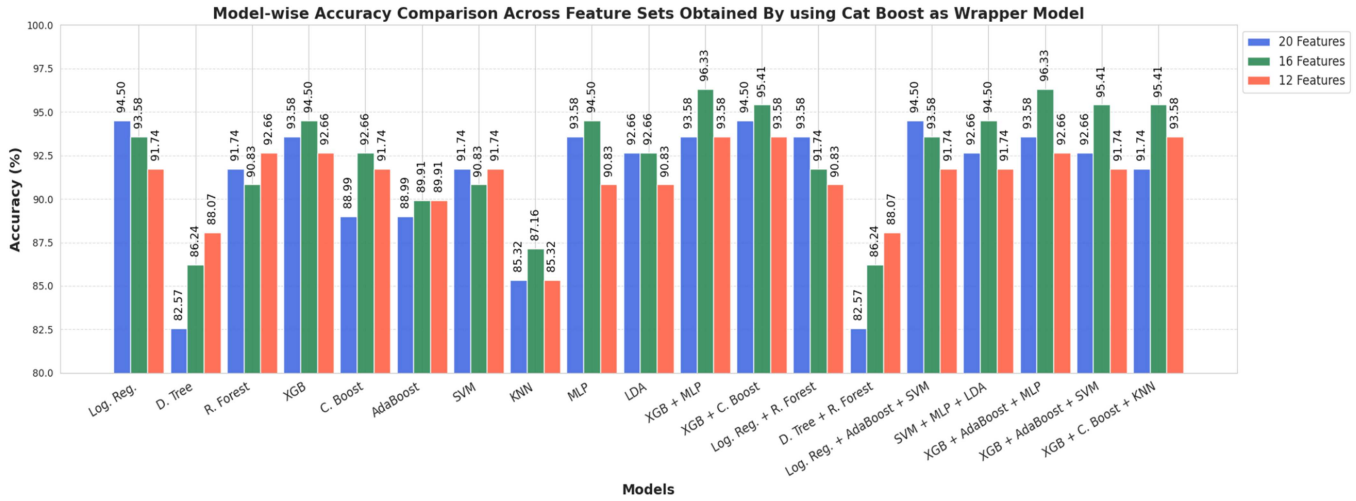


FIGURE 9 | Accuracy of different model using three feature set selected at final stage using RFE + CB.

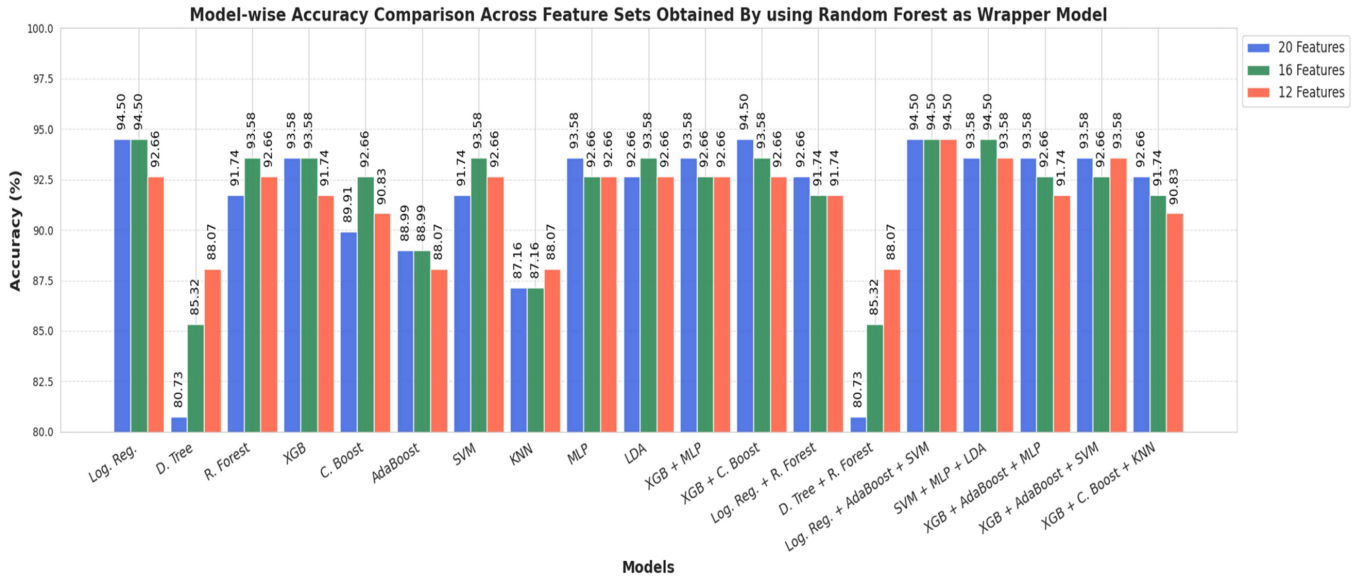


FIGURE 10 | Accuracy of different model using three feature set selected at final stage using RFE + RF.

made it at four and three circumstances, respectively. This shows how vital CB is when it comes to selection traits that provide better predictability. Table 11 well represents the higher accuracy of CB and its higher overall performance.

Performance analysis of the envisioned method versus existing ML approaches: To assess the power of the proposed 16-feature set, it was compared to baseline models and ensemble techniques. Model evaluation was conducted on 16 models using the selected features, which were measured with key performance indicators such as accuracy, recall, precision, and F1-score, as shown in Tables 12 and 13. The results are indisputable, demonstrating that hybrid ensemble models have outperformed individual machine learning models in almost every case. Best performer was the XGB + MLP ensemble with a 96.33% accuracy, 96.35% precision, and 96.30% F1-score. This greatly exceeds the results of classical models such as LR (93.58%) and RF (90.83%). Other hybrid approaches, including XGB + AB + MLP and XGB + CB, also achieved good performance with the accuracy of 96.33% and 95.41%, respectively.

However, baseline models did not perform as well. Traditional classifiers, like KNN (87.16%) and DT (86.24%), achieved low performance, probably due to their sensitivity to irrelevant attributes and their inability to learn complex correlations. More complex models, such as XGB (94.50%) and LR (93.58%), still performed competitively in practice, showing the strength of the feature set that we selected. SVM gave a relatively good accuracy (90.83%) but had higher precision (91.31%) than recall (90.83%), meaning it was designed to avoid false positives. In contrast, AB's performance remained relatively uniform across classes, resulting in almost equal precision and recall values of 89.91%.

The suggested method's true strength is its two-step feature selection process, which, when paired with hybrid ensembles, greatly improves classification performance and resilience. Development of a new hybrid model that incorporates MLP in addition to XGB led to increased accuracy of 3.75% (94.50%–96.33%) compared with standalone XGB. Furthermore, the two-level selection has been effective in reducing risks of

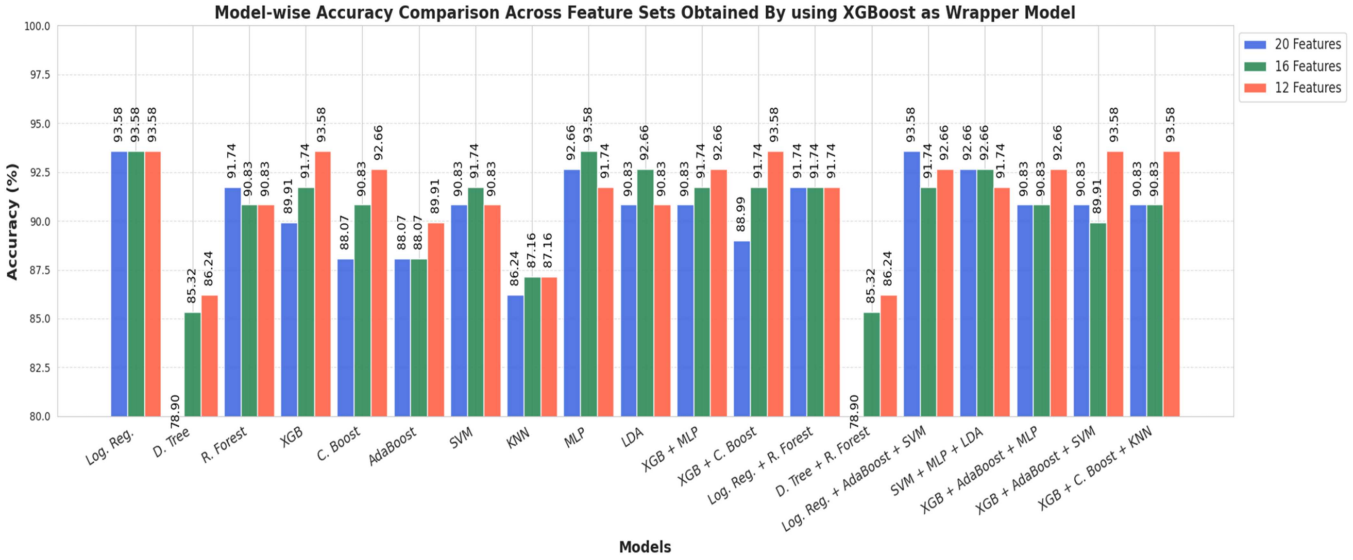


FIGURE 11 | Accuracy of different model using three feature set selected at final stage using RFE + XGB.

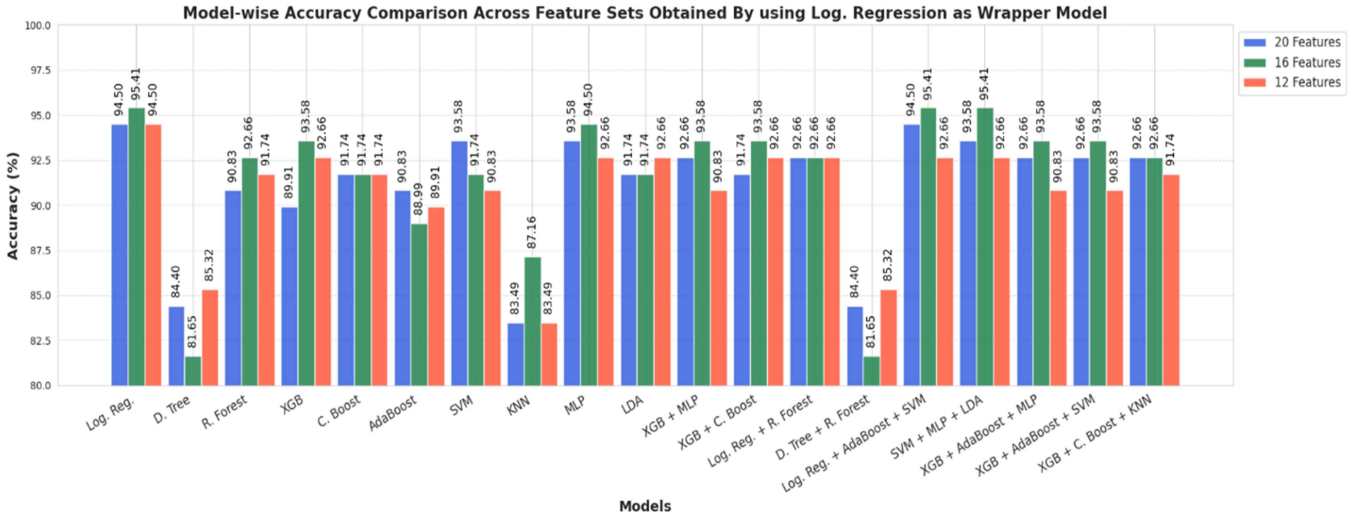


FIGURE 12 | Accuracy of different model using three feature set selected at final stage using RFE + LR.

overfitting and better maintaining the balance between precision and recall through such a broad modeling approach than traditional, single-stage RFE. For classification problems, this also implies that the methodology is a powerful feature selection technique as it is both effective and very efficient, thereby providing generalization and prediction accuracy improvements.

To further assess potential overfitting, both quantitative and graphical evaluations were performed. As shown in Figure 13 the training accuracy remained consistently high 99%, while CV accuracy improved steadily with increasing training size, stabilizing near 0.92. This convergence trend indicates that the model continues to generalize well as more data are introduced, reflecting low overfitting risk. Additionally, Table 14 compares overfitting behavior across models with and without the proposed feature selection approach. It can be observed that, after applying the two-stage feature selection, overfitting margins were substantially reduced for most models. For instance, DT and CB improved from 17.43% and 10.09% to 13.76% and 7.34%, respectively, while the XGB + MLP ensemble decreased from 7.57% to 2.98%. These results demonstrate that the proposed Chi-Square + RFE-based feature selection not only improves predictive performance but also significantly mitigates overfitting, enhancing the model's reliability and stability on unseen data.

4.4 | CV Analysis

To rigorously evaluate the generalizability of the proposed models, a 10-fold stratified CV process was implemented. This approach guarantees the consistency of model performance across a variety of subsets of the data, which is a critical need in clinical diagnostic use. The summarized results are represented in Table 15 and include an analysis of individual and ensemble setups. The CB model was used as a competitive base with a mean accuracy of 92.99% with a standard deviation of 1.68%. The XGB + MLP Soft-Voting model that was proposed as the principal ensemble had a consistent high-performance with a mean accuracy of 92.04% and an F1-score 92.04% with the

TABLE 10 | Statistical analysis data for various ML models with the most important four features.

Feature	Follicle (R)								Cycle (R/I)							
Classifier	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	3.91	0.25	0.55	324.00	0.26	10.64	0.25	NaN	−0.02	0.02	0.07	50.00	0.05	−4.52	0.03	NaN
Y-correlation	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65	−0.18	−0.18	−0.18	−0.18	−0.18	−0.18	−0.18	−0.18
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y1-mean	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.54	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38
Y0-mean	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.43
All mean	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.41
All Std	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.22	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12

Feature	Follicle (L)								Hair growth (Y/N)							
Classifier	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	2.43	0.08	0.03	240.00	0.11	5.22	NaN	0.10	0.82	0.06	0.03	116.00	0.03	2.00	0.12	NaN
Y-correlation	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.60	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y1-mean	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
Y0-mean	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
All mean	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
All Std	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.19	0.45	0.45	0.45	0.45	0.45	0.45	0.45	0.45

Feature	Weight (kg)								RR (breaths/min)							
Classifier	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	−0.07	0.11	0.02	172.00	0.04	−0.31	0.03	NaN	−0.15	0.00	0.00	47.00	0.01	2.01	0.02	NaN
Y-correlation	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39
Y1-mean	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.42	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28
Y0-mean	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.35	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
All mean	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.37	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
All Std	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14

Feature	Cycle length (days)								Marriage status (years)							
Classifier	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	−0.02	0.02	0.07	50.00	0.05	−4.52	0.03	−0.18	−1.80	0.10	0.00	187.00	0.03	−1.57	0.02	NaN

(Continues)

TABLE 10 | (Continued)

Feature / Classifier	Cycle length (days)								Marriage status (years)							
	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
Y-correlation	−0.18	−0.18	−0.18	−0.18	−0.18	−0.18	−0.18	0.00	−0.11	−0.11	−0.11	−0.11	−0.11	−0.11	−0.11	−0.11
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.38	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Y1-mean	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.43	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23
Y0-mean	0.43	0.43	0.43	0.43	0.43	0.43	0.43	0.41	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
All mean	0.41	0.41	0.41	0.41	0.41	0.41	0.41	0.12	0.26	0.26	0.26	0.26	0.26	0.26	0.26	0.26
All Std	0.12	0.12	0.12	0.12	0.12	0.12	0.12	NaN	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16

Feature / Classifier	FSH (mIU/mL)								Avg. *F* size (L) (mm)							
	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	0.00	0.04	0.04	260.00	0.03	−112.6	0.01	NaN	0.24	0.03	0.02	109.00	0.04	1.66	0.04	NaN
Y-correlation	−0.03	−0.03	−0.03	−0.03	−0.03	−0.03	−0.03	−0.03	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
p	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y1-mean	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.65
Y0-mean	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.61	0.61	0.61	0.61	0.61	0.61	0.61	0.61
All mean	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63
All Std	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15

Feature / Classifier	Avg. *F* size (R) (mm)								AMH (ng/mL)							
	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	0.76	0.14	0.04	122.00	0.04	0.80	0.04	NaN	0.03	0.11	0.06	341.00	0.06	−2.12	0.02	NaN
Y-correlation	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
p	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.13	0.13	0.13	0.13	0.13	0.13	0.13	0.13
Y1-mean	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.66	0.53	0.53	0.53	0.53	0.53	0.53	0.53	0.53
Y0-mean	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.63	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49
All mean	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.64	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
All Std	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.14	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29

Feature / Classifier	Weight gain (Y/N)								Skin darkening (Y/N)							
	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
β	0.99	0.06	0.10	72.00	0.08	2.11	0.06	NaN	0.72	0.05	0.00	72.00	0.08	2.19	0.10	NaN
Y-correlation	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.44	0.48	0.48	0.48	0.48	0.48	0.48	0.48	0.48
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

(Continues)

TABLE 10 | (Continued)

Feature Classifier	Weight gain (Y/N)							Skin darkening (Y/N)						
	LR	AB	DT	LGBM	RF	LDA	XGB	MLP	LR	AB	DT	LGBM	RF	LDA	XGB	MLP
Y1-mean	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.68	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62
Y0-mean	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.15	0.15	0.15	0.15	0.15	0.15	0.15	0.15
All mean	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.31	0.31	0.31	0.31	0.31	0.31	0.31	0.31
All Std	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.46	0.46	0.46	0.46	0.46	0.46	0.46	0.46
Feature Classifier	**Pimples (Y/N)**								**Fast food (Y/N)**							
	LR	**AB**	**DT**	**LGBM**	**RF**	**LDA**	**XGB**	**MLP**	**LR**	**AB**	**DT**	**LGBM**	**RF**	**LDA**	**XGB**	**MLP**
β	1.03	0.03	0.02	84.00	0.02	1.26	0.04	NaN	0.33	0.03	0.00	69.00	0.05	0.76	0.04	NaN
Y-correlation	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38
p	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Y1-mean	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.69	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79
Y0-mean	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.39	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38
All mean	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.49	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52
All Std	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50

standard deviation of 2.58% and 2.59%, respectively. Though the original CB system produced favorable results, the XGB + MLP ensemble was selected as the center system due to the gradient-boosting nature of XGB and utilizing the deep feature-extraction ability of MLP. This architectural synergy guarantees stability with respect to various clinical indicators, which maintain high levels of precision and recall in even 10-fold validation.

4.4.1 | *Computational Complexity Analysis*

The computational complexity of the proposed two-level hybrid feature selection framework was analyzed both theoretically and empirically. The overall cost is primarily determined by the number of samples (n), the total number of features (m), and the complexity of the CB model used within the RFE stage.

In the first stage, Chi-Square filtering independently evaluates the statistical dependence between each feature and the target variable, resulting in a time complexity of Big-O ($m \times n$). In the second stage, the RFE procedure iteratively trains CB while removing the least important features, leading to an approximate complexity of Big-O($(p - k) \times n \times t \times \log n$), where p is the number of features retained after the Chi-Square stage, k is the final number of selected features, and t denotes the number of trees in CB. By applying RFE on a smaller subset ($p \ll m$), the proposed two-level framework substantially reduces the overall computational burden compared to traditional single-stage RFE.

The two-level selection significantly enhances computational efficiency while maintaining model stability and reducing overfitting risks, as evidenced by improved generalization performance across the evaluated models shown in Table 16.

By using a Linear SVM model without XAI approaches, Adla et al. [14] were able to attain an accuracy of 91.60%, with a precision of 93.67% and a recall of 80.6%. In a different investigation, Zigarelli et al. [17] combined CB with SHAP explanations, yielding 90.7% recall, 96.0% precision, and 90.10% accuracy. Using RF without XAI techniques, Denny et al. [18] reported an accuracy of 89.02%; however, their model had a comparatively poor recall of 74.19%. In a similar vein, Bharati et al. [21] used RFLR and obtained a 90% recall and 91.01% accuracy. A 94% accuracy rate was attained by Rahman et al. [22] using a mix of AB and RF. Meanwhile, Lu et al. [43] and Shannon et al. [44] obtained lower accuracies of 87% and 82% using DT and XGB methods, respectively. Suha and Islam [45] demonstrated improved performance with a 95.7% accuracy using GB. In contrast, the proposed method, which leverages a combination of XGB and MLP models along with SHAP and LIME for explainability, attained the highest performance, achieving an accuracy of 96.33%, a precision of 96.35%, and a recall of 96.33%. These findings demonstrate that the proposed approach surpasses existing methods, providing a more reliable and interpretable solution within the scope of this study (Table 17).

4.5 | XAI

XAI aims to provide clear and interpretable explanations of how ML models arrive at their predictions and to alleviate the

TABLE 11 | Key findings ML model versus final feature set (16 features contains).

Matrices	16 Features and different wrapper models used in RFE			
	LR (%)	XGB (%)	RF (%)	CB (%)
1st Rank accuracy	95.41	93.58	94.50	96.33
2nd Rank accuracy	95.41	93.58	94.50	96.33
3rd Rank accuracy	94.50	92.66	94.50	95.41
4th Rank accuracy	94.50	92.66	93.50	95.41
Minimum accuracy	81.65	85.32	85.32	86.24
Average accuracy	91.65	90.73	91.89	92.47

TABLE 12 | Model's performance of different ML models using the proposed feature set.

Model	Performance using final feature set			
	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
LR	93.58	93.64	93.58	93.60
DT	86.24	86.35	86.24	86.29
RF	90.83	90.83	90.83	90.83
XGB	94.50	94.49	94.50	94.45
CB	92.66	92.62	92.66	92.61
AB	89.91	89.90	89.91	89.94
SVM	90.83	91.31	90.83	90.94
KNN	87.16	87. 16	87.16	87.16
MLP	94.50	94.50	94.50	94.50
LDA	92.66	92.66	92.66	92.66

black-box nature of complex machine learning algorithms. By using methods such as intrinsic interpretable models and post-hoc explanation methods (LIME, SHAP), XAI allows for a deeper explanation, for instance. This transparency enhances trust and confidence in vital applications by removing potential biases, ensuring accountability, and validating model performance.

4.5.1 | *SHAP Values*

This study explores several ensemble models to improve predictive performance and finds that the optimal combination of XGB and MLP achieves the best accuracy. Since the ensemble model is more complex we also form a SHAP analysis specifically for this model to understand the features that contribute both models and interpret the features that had the most effect on prediction. SHAP summary plots are shown in Figure 14 for PCOS-positive cases and Figure 15 for PCOS-negative cases. Machine learning models can also be interpreted using SHAP values, which provide a significance value to each feature for a specific prediction.

The *y*-axis for both plots ranks features by how useful they are for predicting the response of the model, with "Follicle (R)" and "Follicle (L)" highlighted as the most influential elements. Other important features are "Cycle (R/I)," "hair growth (Y/N)," and "weight gain (Y/N)." The *x*-axis denotes SHAP values, which represent the impact of each attribute on the prediction. Negative SHAP values show a negative correlation, and positive SHAP values indicate a positive correlation to the outcome. The feature values are denoted by the gradient of colors from red to blue, where blue tells us lesser feature values, while red indicates higher ones. This is because higher (left and right) follicle measurements in Figure 14 are highly correlated with positive SHAP values, indicating that they strongly influence the model's ability to identify PCOS-positive patients. In contrast, negative values of SHAP can be observed for the reduced counts of follicles in Figure 15, confirming their role and importance in making predictions regarding PCOS-negative outcomes. "The FSH (mIU/mL)" in Figure 14 and "Fast food (Y/N)," "Skin darkening (Y/N)" in Figure 15 are included to predict outcomes as well, even though the effect differs.

These graphs demonstrate how various levels of features impact the model's predictions, providing valuable insight into significant contributors to PCOS outcomes. SHAP values allowed users to find out more about the underlying factors that were impacting the model's predictions, leading to machine learning applications that are easier to understand and interpret.

4.5.2 | *LIME*

To improve the accuracy of ML model predictions, particularly in important fields such as healthcare, XAI techniques have gained great popularity. Given that many machine learning models are black-boxed, interpretability is as important as getting high performance. To foster confidence, transparency, and robustness in AI-based diagnostics, XAI techniques help disentangle a model's decision processes. To improve prediction performance, this study experiments with several kinds of ensemble models; the best accuracy is obtained by combining the XGB and MLP. The LIME analysis, which has been applied to this model, offers some insight into the features that are crucial for prediction and allows for some interpretation of the model because LIME estimates the model around the prediction level.

In this study, we examine and explain the decisions of our proposed model for predicting PCOS utilizing LIME. Because it describes the classifier's decision-making method for individual samples rather than the entire dataset, LIME becomes particularly useful for local interpretability. Through feature emphasis value assignment, LIME offers information about which traits are most important for a given prediction.

TABLE 13 | Model's performance of stacking ensemble classifiers using the proposed feature set.

Model	Performance using final feature set			
	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
XGB + MLP	96.33	96.35	96.33	96.30
XGB + CB	95.41	95.47	95.41	95.36
LR + RF	91.74	91.70	91.74	91.71
DT + RF	86.24	86.35	86.24	86.29
LR + AB + SVM	93.58	93.85	93.58	93.64
SVM + MLP + LDA	94.50	94.50	94.50	94.50
XGB + AB + MLP	96.33	96.35	96.33	96.30
XGB + AB + SVM	95.41	95.47	95.41	95.36
XGB + CB + KNN	95.41	95.47	95.41	95.36

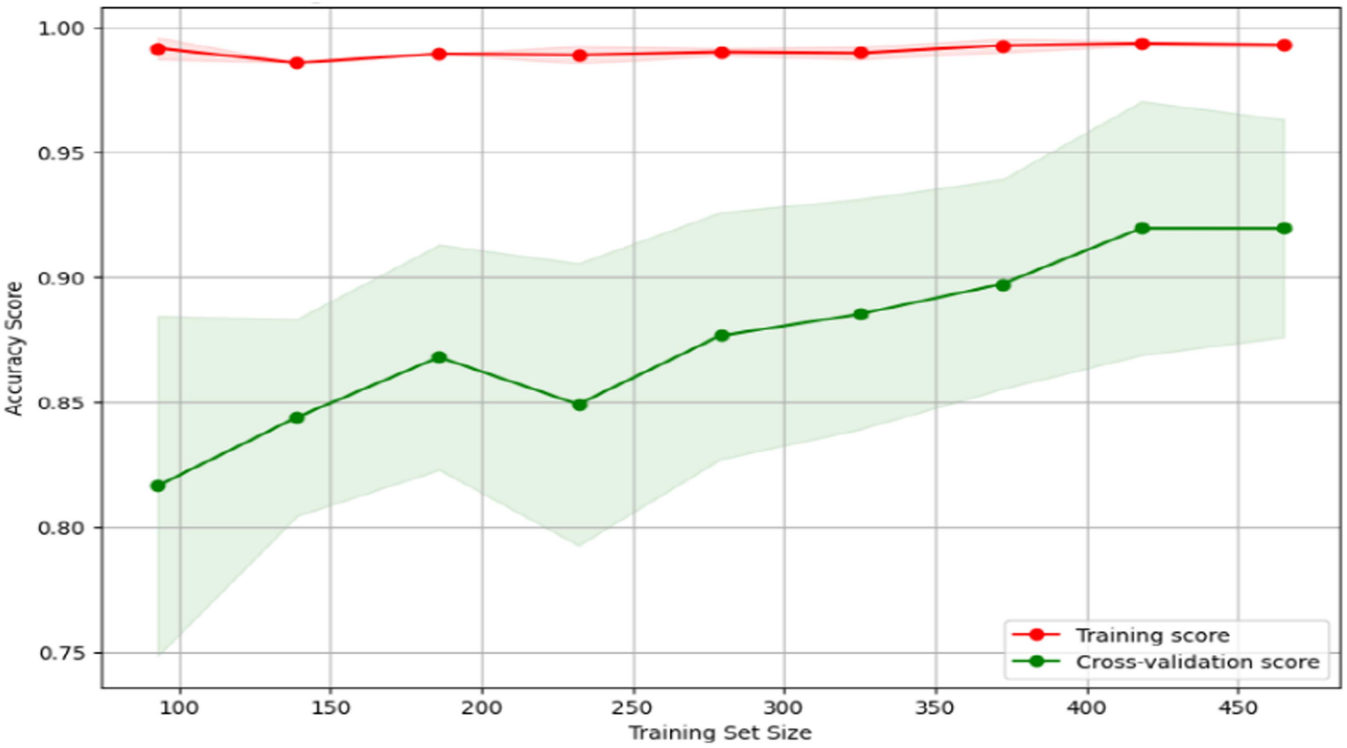


FIGURE 13 | Learning curve for XGB + MLP ensemble model.

For implementation, we utilize the "lime" Python library, which allows for regression and classification analysis. We use the classification mode, which generates a bar plot of the importance of each feature, color-coded as red for features that influence the predicted class positively and as blue for features that contribute negatively, because our model predicts a categorical variable (PCOS/non-PCOS). Orange bars indicate support for a PCOS categorization, and blue bars indicate support for a non-PCOS classification. The length of each bar corresponds to how much the feature contributes to the prediction. Figure 16 shows the LIME for a non-PCOS example, where class 0 (non-PCOS) is predicted with 100% probability. The most relevant features for this decision are Hair Growth (Y/N: −0.72, 6%), Follicle Number (Right: −1.50, 39%), and Follicle Number (Left: −1.42, 21%). Disapplying a diagnosis of PCOS is highly encouraged by poor results that reveal a low number of follicles, normal menstrual periods, and few symptoms. PCOS is weakly affected by features such as AMH (ng/mL: −0.63, contributing 2%) and FSH (mIU/mL: −0.04, contributing 2%), such that they are not sufficient to affect the model's final result.

In a PCOS-positive case, the model shows a 100% probability of PCOS as depicted in Figure 17. The three most important features were Follicle Number (Right: 1.68, weight 25%), Follicle Number (Left: 1.16, weight 20%), and Cycle Regularity (R/I: 1.34, weight 12%). All of these factors clearly demonstrate PCOS characteristics such as irregular periods and an increased number of follicles. The non-PCOS label acquired at a selection of the other features, such as Hair Growth (Y/N: −0.72, contributing 5%), is outmatched by stronger PCOS markers such as AMH (ng/mL: 0.79, contributing 4%),

TABLE 14 | Overfitting analysis across classifiers with and without feature selection.

	Overfitting analysis with selected features			Overfitting analysis without feature selection		
Model	**Train accuracy (%)**	**Test accuracy (%)**	**Overfitting (%)**	**Train accuracy (%)**	**Test accuracy (%)**	**Overfitting (%)**
LR	91.92	93.58	−1.65	91.41	94.50	−3.09
DT	100.00	86.24	13.76	100.00	82.57	17.43
RF	100.00	90.83	9.17	100.00	90.83	9.17
AB	92.61	89.91	2.70	95.02	87.16	7.86
XGB	100.00	94.50	5.50	100.00	90.83	9.17
CB	100.00	92.66	7.34	100.00	89.91	10.09
SVM	94.50	90.83	3.68	95.02	93.58	1.44
KNN	91.07	87.16	3.91	90.38	87.16	3.22
GNB	85.40	80.73	4.66	86.08	82.57	3.51
MLP	92.78	94.50	−1.71	95.02	92.66	2.36
LDA	92.10	92.66	−0.56	92.78	93.58	−0.79
XGB + MLP	99.31	96.33	2.98	99.31	91.74	7.57

TABLE 15 | Performance metrics of individual and ensemble machine learning models using 10-fold stratified cross-validation.

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 score (%)**
CB	92.99 ± 1.68	93.21 ± 1.62	92.99 ± 1.68	92.98 ± 1.68
XGB + CB + KNN (Soft Voting)	92.44 ± 2.15	92.76 ± 1.90	92.44 ± 2.15	92.43 ± 2.17
SVM + MLP + LDA (Soft Voting)	92.44 ± 1.66	92.60 ± 1.57	92.44 ± 1.66	92.43 ± 1.67
LR + AdaBoost + SVM (Soft Voting)	92.30 ± 2.24	92.49 ± 2.14	92.30 ± 2.24	92.29 ± 2.25
XGB	92.17 ± 3.13	92.46 ± 2.95	92.17 ± 3.13	92.16 ± 3.14
XGB + CB (Soft Voting)	92.17 ± 2.46	92.49 ± 2.26	92.17 ± 2.46	92.16 ± 2.47
XGB + MLP (Soft Voting)	92.04 ± 2.58	92.24 ± 2.50	92.04 ± 2.58	92.03 ± 2.59
XGB + AdaBoost + MLP (Soft Voting)	92.04 ± 2.43	92.29 ± 2.25	92.04 ± 2.43	92.02 ± 2.44
XGB + AdaBoost + SVM (Soft Voting)	91.89 ± 2.49	92.15 ± 2.33	91.89 ± 2.49	91.88 ± 2.50
MLP	91.76 ± 1.36	91.85 ± 1.30	91.76 ± 1.36	91.75 ± 1.37
LR	91.75 ± 2.54	92.04 ± 2.37	91.75 ± 2.54	91.74 ± 2.56
SVM	91.34 ± 2.23	91.55 ± 2.36	91.34 ± 2.23	91.34 ± 2.22
AdaBoost	91.07 ± 3.56	91.26 ± 3.61	91.07 ± 3.56	91.06 ± 3.57
LDA	90.25 ± 2.25	90.54 ± 2.00	90.25 ± 2.25	90.23 ± 2.27
KNN	87.10 ± 3.34	87.73 ± 3.18	87.10 ± 3.34	87.03 ± 3.39
DT	86.96 ± 3.46	87.24 ± 3.22	86.96 ± 3.46	86.93 ± 3.49

TABLE 16 | Computational complexity on different stage of proposed method.

Stage	**Operation**	**Theoretical complexity**
Chi-Square filtering	Univariate dependence testing	Big-O $(m \times n)$
RFE with CatBoost	Iterative elimination and retraining	Big-O $((p - k) \times n \times t \times \log n)$
Model training (Final Ensemble)	Training and evaluation	Big-O $(n \times t \times \log n)$

Skin Darkening (Y/N: 1.28, contributing 3%), and Weight (kg: −0.59, contributing 3%).

These findings demonstrate that LIME improves interpretation with a PCOS prediction model by explicitly weighting clinical characteristics. The visuals establish the model's trustworthiness in real-world applications, as it highly correlates with well-known medical characteristics of PCOS. In an effort to build more transparent and efficient AI-driven diagnostic tools, we employ XAI to ensure that the medical practitioners can understand, trust, and refine the model.

TABLE 17 | A comparison analysis between the proposed method and recent works.

Title	Method	XAI	Accuracy (%)	Precision (%)	Recall (%)	F1 score (%)
Adla et al. [14]	Linear SVM	No	91.60	93.67	80.60	86.60
Zigarelli et al. [17]	CB	SHAP	90.10	96.00	90.70	93.20
Denny et al. [18]	RF	No	89.02	95.83	74.19	41.82
Bharati et al. [21]	RFLR	No	91.01	89.90	90.00	89.95
Rahman et al. [22]	AB and RF	No	94.00	93.00	94.00	93.00
Lu et al. [43]	DT	No	87.00	85.00	84.00	84.00
Suha and Islam [45]	GB	No	95.70	95.00	96.00	96.00
Shannon et al. [44]	XGB	No	82.00	81.00	80.00	80.00
Proposed method	**XGB, MLP**	**SHAP, LIME**	**96.33**	**96.35**	**96.33**	**96.30**

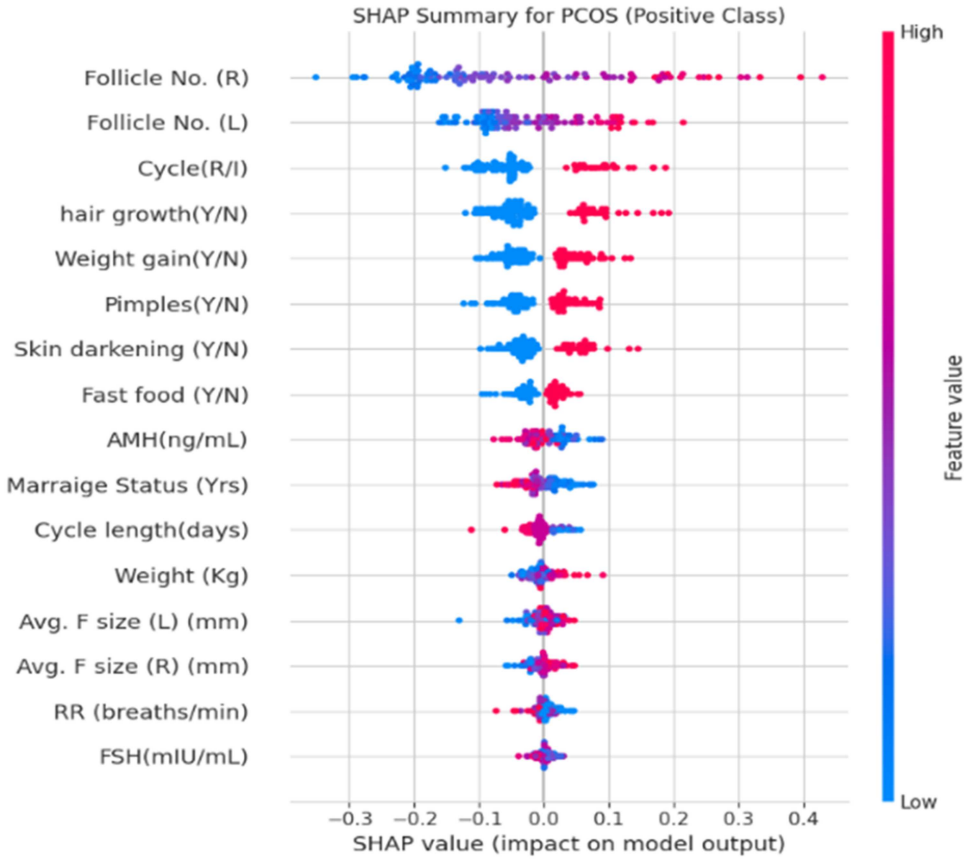


FIGURE 14 | SHAP summary plots for PCOS positive case.

4.6 | Clinical Applicability and Study Limitations

Although the suggested two-stage feature selection and hybrid ensemble model are shown to be highly predictive, the transition between a computational model and real-world clinical implementation should be taken into consideration. This framework is designed to be a decision-support tool at the early stage as opposed to a decision-making tool in its own right. An effective implementation would require an easy integration with the current hospital Electronic Health Record systems or a secure Web-based interface. In this process, doctors would be able to enter general patient vital signs and initial laboratory findings. When the model indicates a high risk of PCOS, clinicians can use the combined SHAP and LIME visual dashboards to immediately understand the physiological explanation behind the prediction. This openness will make the output of the AI consistent with the accepted medical knowledge and can help the doctors prioritize the patients to receive specific, expensive follow-up diagnostics, including pelvic ultrasounds or a specific hormonal panel.

In spite of such promising applications, this study has some significant limitations. The framework was trained and tested on a publicly available dataset of 541 patients. According to the literature available, this data set is mostly based on a particular geographic and demographic group of people in India. Since the

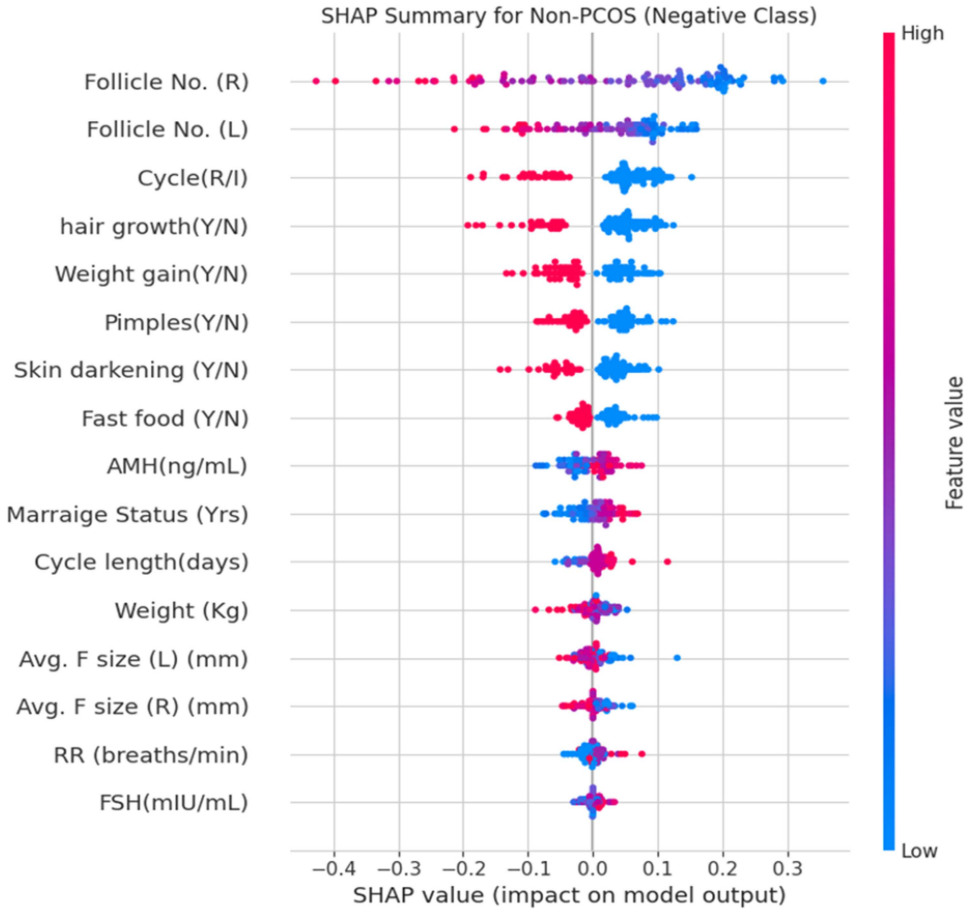


FIGURE 15 | SHAP summary plots for PCOS negative case.

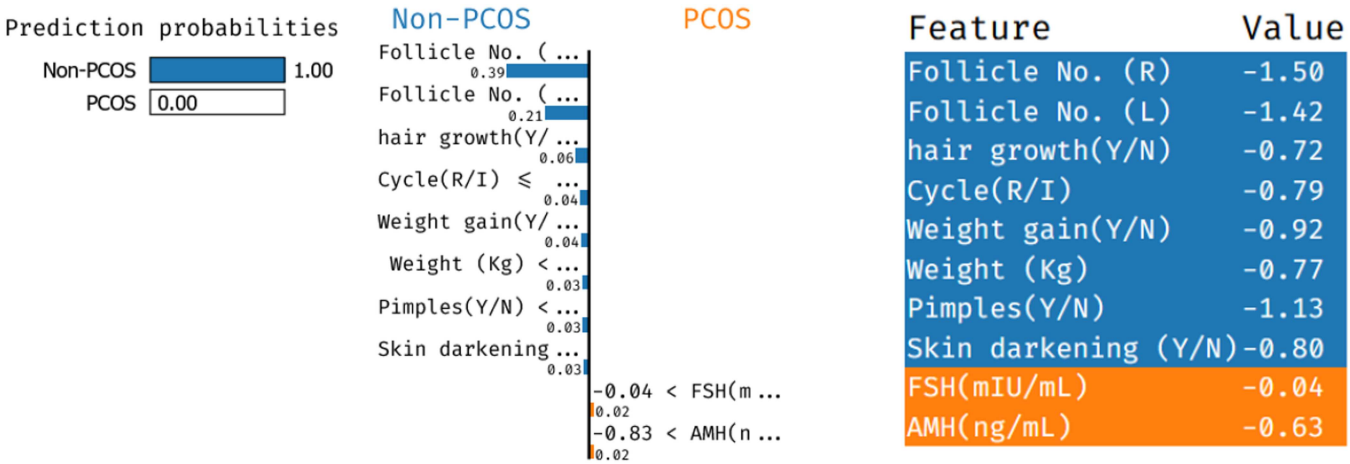


FIGURE 16 | LIME for a Non-PCOS case.

presentation of PCOS, baseline metabolic parameters, and the presence of symptoms (including hirsutism or weight gain) may differ greatly across ethnic, racial, and socioeconomic groups, the existing dataset is bound to introduce possible geographic and demographic bias. As a result, the applicability of the proposed XGB + MLP model to a worldwide population is limited now. In order to adopt this framework safely and fairly in a variety of clinical settings, it is first necessary that it be externally validated and recalibrated with large, multi-center, and multi-ethnic cohorts to provide algorithmic fairness and strong real-world generalizability.

5 | Conclusion

Patients with PCOS may have infertility if the problem is not detected early enough. The PCOS patient rise rate is larger than previous years' records due to difficulties in early identification.



FIGURE 17 | LIME for a PCOS positive case.

By employing an advanced machine learning pipeline, our strategy facilitates early identification of PCOS risk with strong and precise predictions. We integrate multiple classifiers, including LR, XGB, DT, RF, CB, AB, SVM, LDA, KNN, and MLP, to capture complementary patterns in the data, and we strengthen the predictive signal through a two-stage feature selection process using Chi-Square followed by RFE. This integrated approach improves accuracy, precision, recall, and F1 score while maintaining interpretability and practical efficiency. In terms of PCOS prediction, we achieved an accuracy of 96.33%, a precision of 96.35%, a recall of 96.33%, and an F1 score of 96.30%, establishing new standards. Information remained interpretable with XAI techniques like SHAP and LIME, enabling transparent clinical decision-making. The findings support faster and more accurate identification of PCOS for patients and clinicians by unifying the strengths of multiple machine learning classifiers in a single robust model based on a compact set of predictors. The principal contribution is a two-stage feature selection strategy using Chi Square and RFE, together with rigorous treatment of feature selection, normalization, imputation for missing data, and a voting ensemble model.

Author Contributions

Md. Rakibul Hasan Efty: conceptualization, methodology, software. **Md. Naymur Rohman:** resources, formal analysis, writing – original draft. **Khandaker Mohammad Mohi Uddin:** supervision, validation, methodology, writing – original draft. **Md. Mahbubur Rahman:** validation, visualization. **Md. Ashraf Uddin:** writing – review and editing, supervision.

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Ethics Statement

This research involved no human subjects. This study adhered to the rules set out by the "Committee for the Purpose of Control and Supervision of Experiments on Animals" (CPCSEA) regarding animal welfare.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request. https://github.com/KMM-Uddin/XAI_PCOS-Prediction.git.

Transparency Statement

The corresponding author affirms that this manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned have been explained.

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